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Renal Circulation and Glomerular Filtration

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CHAPTER OUTLINE

KEY POINTS

- Renal blood flow (RBF) is 20% of the cardiac output, and RBF and glomerular filtration rate (GFR) increase with protein intake in men and women. At any given level of protein intake, GFR is greater in younger than older adults and greater in men than women.
- The decline in hydraulic pressure between the renal artery and glomerular capillaries is greatest along the afferent arteriole, whereas 70% of the decline in postglomerular hydraulic pressure occurs at the efferent arteriole. The last 50 to 150 µm of the afferent arteriole and the first early part of the efferent arteriole (first 50–150 µm) provide most of the preglomerular and postglomerular resistances.
- Glomerular capillary blood pressure, P_{GC}, declines only slightly within the capillary network, thus maintaining the transcapillary hydraulic pressure gradient ($\Delta P = P_{GC}$ minus Bowman space pressure, P_{BS}) at a relatively constant rate. However, glomerular capillary protein oncotic pressure (π_{GC}) rises along the length of the glomerular capillaries as protein-free fluid is filtered into the Bowman space, thus reducing the net filtration pressure.
- The barrier to filtration of fluid and macromolecules includes the glycocalyx lining the endothelial cells, the fenestrations of the endothelial layer of the glomerular capillaries, the layers of the glomerular basement membrane, the filtration slits between the podocytes surrounding the capillaries, and the filtration slit diaphragm connecting adjacent foot processes. Breakdown or injury in any of the restrictive barriers may reduce GFR but can also lead to increased passage of albumin and other proteins into the Bowman space.
- The high resistance of the afferent and efferent arterioles leads to a large drop in vascular hydraulic pressure before the peritubular capillaries so that peritubular capillary pressure is much lower than glomerular capillary pressure (15–20 mm Hg). The oncotic pressure of fluid entering the peritubular capillaries is elevated because of the filtration of protein-free fluid out of the glomerular capillaries. The net balance between the transcapillary oncotic and hydraulic pressure gradients thus favors entry of the fluid reabsorbed by the tubules into the peritubular capillaries.
- RBF and GFR are autoregulated, indicating that the principal resistance adjustments are localized at the preglomerular vasculature. The autoregulation mechanisms maintain the intrarenal hemodynamic environment in balance with the metabolically determined tubular transport function.
- The myogenic and tubuloglomerular feedback mechanisms are responsible for the autoregulatory responses. The myogenic mechanism refers to the ability of arterial smooth muscle to contract and relax in response to increases and decreases in vascular wall tension. The tubuloglomerular feedback mechanism provides signals from the macula densa to the afferent arteriole. Paracrine mediators include ATP, adenosine, nitric oxide, and arachidonic acid metabolites.

KEY POINTS—cont'd

- Overall renal hemodynamic function is regulated by complex interactions between extrinsic and intrinsic mechanisms, including the sympathetic nervous system, circulating vasoactive factors, nitric oxide, intrarenal angiotensin II, arachidonic acid metabolites, purinergic factors, and other paracrine systems, which exert direct effects and modulate the sensitivity of the tubuloglomerular feedback mechanism.

The kidneys are unique in having three distinct microvascular networks identified as the glomerular capillary microcirculation, the cortical peritubular capillary microcirculation, and the medullary microcirculation. These circulations have specialized functions that allow filtration of a large volume of fluid at the glomerular capillaries, the consequent reabsorption of most of the filtrate back into the circulation, the secretion of numerous substances from capillary plasma to the tubular fluid, and the establishment of a medullary environment with a high interstitial osmolality. These vascular beds provide the renal cells and tissues with oxygen and nutrients, as well as regulate the hemodynamic environment to achieve their designated functions. Under resting conditions, blood flow to the kidneys represents approximately 20% of cardiac output in humans, even though these organs constitute less than 1% of body mass. Renal blood flow (RBF), approximately 400 mL/100 g of tissue per minute, is significantly greater than that observed in other vascular beds such as the heart, liver, and brain.[1,2] From this enormous blood flow (1.0–1.2 L/min), approximately 20% of the plasma flow is filtered and becomes the glomerular filtrate, but most is reabsorbed back into the vasculature, leaving only a small fraction, about 1%, to be excreted as urine. Although the metabolic energy requirements of tubular transport processes are relatively high, the renal arteriovenous O_2 difference reveals that blood flow far exceeds that needed for metabolic demands. In fact, the high blood flow is essential to provide the appropriate hemodynamic environments necessary for the filtration at the glomeruli and the secretion from and reabsorption into the postglomerular capillaries.[3,4]

In Chapter 2, the gross anatomy of the kidney and arrangement of tubular segments are described in detail. In this chapter, we consider the intrarenal organization of the discrete microcirculatory networks, as shown in Fig. 3.1 and Fig. 3.2. We also consider the differences in regional blood flows and how the structure of the microcirculation contributes to the regulation of the intrarenal hemodynamic environment, thus maintaining appropriate levels of RBF and GFR.

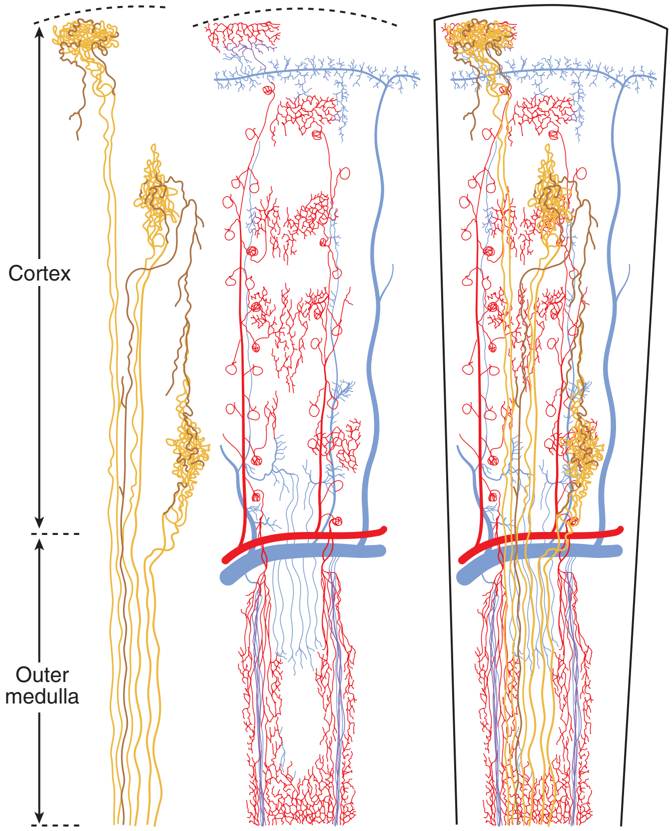

Fig. 3.1 renal vasculature and tubule organization. *Left,* Three nephrons are shown without accompanying vascular structures. Vascular structures are shown in the *central portion* of the figure. *Right,* Vascular and tubular structures are superimposed. Configurations of tubular segments were generalized from patterns found by silicone rubber injections. For clarity, more distal parts of the nephron are shown in deeper colors. Arterial components of the vascular system are shown in *red,* venous components in *blue*. Only representative venous connections are shown. (From Beeuwkes R III, Bonventre JV. Tubular organization and vascular tubular relations in the dog kidney. *Am J Physiol.* 1975;229:695–713.)

RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

Historically, GFR and renal plasma flow (RPF) have been estimated using the clearance of inulin for the determination of GFR and of *p*-aminohippuric acid (PAH) for the determinations of RPF, from which RBF can be calculated from RPF and hematocrit.[5] Whereas inulin is only filtered across the glomerular capillaries, PAH is both filtered at the glomerulus and actively secreted by the tubules. This results in the renal extraction of 80% to 90% of PAH from the blood. PAH is not completely extracted from the blood because of flow through regions of the kidney, in particular the medulla, that does not perfuse proximal tubule segments, where secretion occurs, and limitations of the secretory process of PAH is in cortical regions. Thus PAH clearance is an approximation often termed “effective” or “estimated” renal plasma flow (ERPF) and provides an estimate of RPF without the need for a renal venous blood sample.[5] However, this estimate of RPF is much less accurate in renal disease because extraction is further reduced by damage to proximal tubule segments or rarefaction of the peritubular

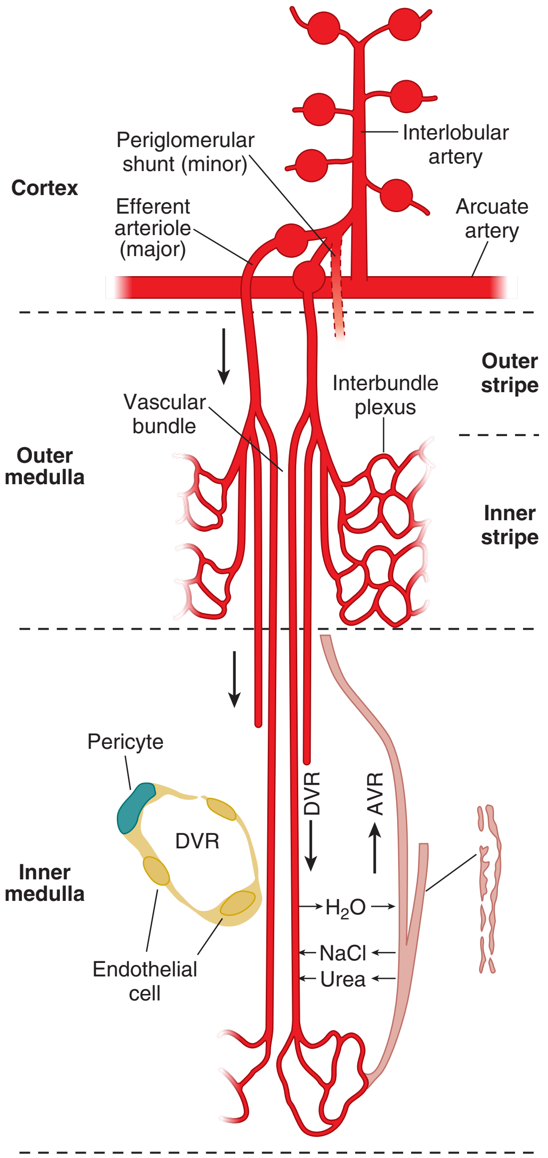

Fig. 3.2 The medullary microcirculation. In the cortex, interlobular arteries arise from the arcuate artery and ascend toward the cortical surface. Cortical and juxtamedullary afferent arterioles leading to glomeruli branch from the interlobular arteries. The blood reaches the medulla through juxtamedullary efferent arterioles; however, a tiny (<1%) fraction may also arise from periglomerular shunt pathways. In the outer medulla, juxtamedullary efferent arterioles in the outer stripe give rise to descending vasa recta *(DVR),* which coalesce to form vascular bundles in the inner stripe. DVRs on the periphery of vascular bundles give rise to the interbundle capillary plexus that surrounds nephron segments—thick ascending limb, collecting duct, long looped thin descending limbs (not shown). DVRs in the center continue across the inner-outer medullary junction to perfuse the inner medulla. Vascular bundles disappear in the inner medulla, and vasa recta become dispersed with nephron segments. Ascending vasa recta *(AVRs)* that arise from the sparse capillary plexus of the inner medulla return to the cortex by passing through outer medullary vascular bundles. *Inset,* DVRs have a continuous endothelium and are surrounded by contractile pericytes. AVRs have fenestrations allowing greater permeability. (From Pallone TL, Zhang Z, Rhinehart K. Physiology of the renal medullary microcirculation. *Am J Physiol*. 2003;284:F253–F266, 2003.)

Table 3.1 Renal Blood Flow, Renal Plasma Flow, Glomerular Filtration Rates, and Filtration Fractions in Healthy Men and Women[a]

Subject(s)	RBF, mL/min	RPF, mL/min	GFR, mL/min	Filtration Fraction
Men	1166	655	127	0.193
Women	940	600	118	0.197
Combined	1165	634	123	.197

[a]Data for RPF based on clearances of Diodrast and *p*-aminohippuric acid (PAH) and for GFR based on clearance of inulin. Clearances are corrected to 1.73 m^2 and suggest lower normalized RBF, RPF, and GFR values in women.
GFR, Glomerular filtration rate; *RBF,* renal blood flow; *RPF,* renal plasma flow.
Data from Smith HW. *The Kidney: Structure and Function in Health and Disease*. New York, NY: Oxford University Press; 1951:544–545.

capillaries involved in PAH secretion.[6] Values taken from a composite of studies in normal human subjects[5] are shown in Table 3.1. As shown, RBF and GFR are lower in women than in men, even when corrected for body surface area. Values for normal subjects vary considerably in different studies, as reflected in Fig. 3.3, which shows data on ERPF and GFR in adult humans from various studies. The data also reveal marked differences in GFR and ERPF between obese and lean subjects. These differences are likely the result of increased food intake and, hence, increased protein consumption.[7] Indeed, as shown in Fig. 3.4, GFR increases with protein intake in men and women and, at any given level of protein intake, GFR is higher in young people (20–50 years of age; mean, 31 years) than older subjects (55–88 years of age; mean, 70 years). Improved methods of RBF measurement include laser Doppler flowmetry, video microscopy, and imaging techniques such as positron emission tomography (PET), high-speed computed tomography (CT), and magnetic resonance imaging (MRI) that have been especially useful in determining differences in regional blood flow.[6,8-12]

MAJOR ARTERIES AND VEINS

Separate renal arteries that branch directly from the abdominal aorta provide blood supply to each kidney. The arteries branch into multiple segmental vessels at a point just before entry into the renal parenchyma and continue to branch in a nonanastomotic manner to supply the glomeruli before entering the postglomerular microcirculation (see Fig. 3.1).[3] Because the branches do not anastomose, complete obstruction of an arterial segmental vessel results in ischemia and infarction of the tissue in its area of distribution. Ligation of individual segmental arteries has frequently been performed in experimental studies to reduce renal mass and produce the remnant kidney model of chronic renal failure.[13,14] Morphologic studies in this model have revealed the presence of ischemic zones adjacent to the totally infarcted areas. These regions contain viable

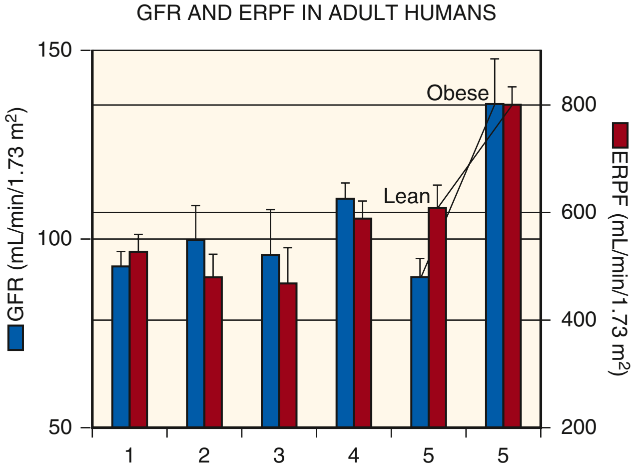

Fig. 3.3 Representative values for glomerular filtration rate *(GFR)* and estimated renal plasma flow *(ERPF)* from five studies in adults. Values from men and women were pooled. Numbers under each set of bars refer to the following studies: 1. Giordano and DeFronzo[525]; 2. Winetz et al[526]; 3. Hostetter[527]; 4. Deen et al[528]; 5. Chagnac and colleagues.[529] For studies 1 through 3 and 5, values were obtained after approximately 12 hours of fasting; subjects in study 4 were allowed food ad lib. For study 5, values from lean subjects (average body mass index [BMI] = 22) were compared with those from obese nondiabetic individuals (BMI >38) after a 10-hour fast but are not corrected for body surface area.

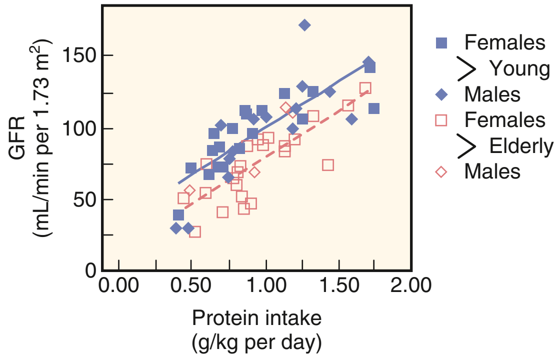

Fig. 3.4 Relationship between protein intake and glomerular filtration rate *(GFR)* determined from creatinine clearances. Data from younger (mean age, 31 years; range, 20–50 years) and older (mean age, 70 years; range, 55–88 years) healthy humans. *Closed symbols,* younger subjects; *open symbols,* older subjects; *squares,* women; *diamonds,* men. (Data from Lew SQ, Bosch P. Effect of diet on creatinine clearance in young and elderly healthy subjects and in patients with renal disease. *J Am Soc Nephrol*. 1991;2:856–865.)

glomeruli that appear shrunken and crowded together, demonstrating that few portions of the renal cortex have partial dual perfusion.[15] The anatomic distribution just described is most common; however, other patterns may occur.[16,17] Secondary renal arteries found in 20% to 30% of normal individuals may result from division of the renal artery at the aorta. These vessels, which most often supply the lower pole,[18] may be the sole arterial supply of some parts of the kidney.[19]

Within the renal sinus of the human kidney, division of the segmental arteries gives rise to the interlobar arteries. These vessels, in turn, give rise to the arcuate arteries, whose several divisions lie at the border between the cortex and medulla. The interlobular arteries branch from the arcuate arteries more or less sharply, usually as a common trunk that divides two to five times as it extends toward the kidney surface[20,21] (see Fig. 3.1). Afferent arterioles leading to glomeruli arise from the interlobular arteries (see Figs. 3.1 and 3.2). Glomeruli are classified according to their position within the cortex as superficial (i.e., near the kidney surface), midcortical, or juxtamedullary, near the corticomedullary border (see Fig. 3.1). The capillary network of each glomerulus originates from the afferent arteriole as it enters into a manifold-like chamber. The glomerular capillaries coalesce into an efferent chamber leading to an efferent arteriole that delivers blood to the postglomerular capillary circulation, forming both the cortical peritubular capillaries and complex medullary capillaries. The arrangement of the medullary microcirculation plays an important role in the process of concentration of urine.

Venous drainage of the peritubular capillaries from the superficial cortex is via superficial cortical veins.[20,22] In the middle and inner cortices, venous drainage is achieved mainly by the interlobular veins. The dense peritubular capillary network surrounding the interlobular vessels drains directly into the interlobular veins through multiple connections, whereas the less dense, long-meshed network of the medullary rays appears to anastomose with the interlobular network and thus drains laterally (see Fig. 3.1). The medullary circulation also shows two different types of drainage. The outer medullary networks typically extend into the medullary rays before joining interlobular veins, whereas the long vascular bundles of the inner medulla (vasa recta) converge abruptly and join the arcuate veins.

Renal lymphatics and their roles in physiology and disease have been largely overlooked, although they are abundant mostly in the cortex rather than the medulla of the normal kidney. Lymphatic vessels originate as blind-ended initial capillaries and can either follow the main arteries and veins toward the hilum or penetrate the capsule to join capsular lymphatics. The physiologic function of renal lymphatics is to drain the interstitial fluid in the cortex. It is largely composed of capillary filtrate but also contains fluid reabsorbed from the tubules. The main factors that contribute to renal lymph formation are interstitial fluid volume, intrarenal venous pressure, and capillary permeability. Elevations in these factors can lead to renal interstitial edema and decreased kidney function in many pathologies (e.g., cardiac failure and inflammatory conditions). Kidney lymphangiogenesis is strongly associated with injury, inflammation, and the progression of fibrosis.[23-25]

OXYGEN CONSUMPTION

Because of the unique juxtaposition of the arteriolar and venular network, much of the abundant oxygen supply to the kidneys diffuses from the arterioles to the venules (Fig. 3.5).[26] The shunting of oxygen, coupled with the high rate of oxygen consumption (~4 μmol/min/g), leaves the oxygen tension (po_2) in the cortex much lower than what would be predicted from the po_2 in renal venous blood. Tissue po_2 values in the cortex border on hypoxia, varying from 40 to 45 mm Hg in the outer and mid cortices and even lower (30 mm Hg) in the deep cortex.[13,27,28] The countercurrent

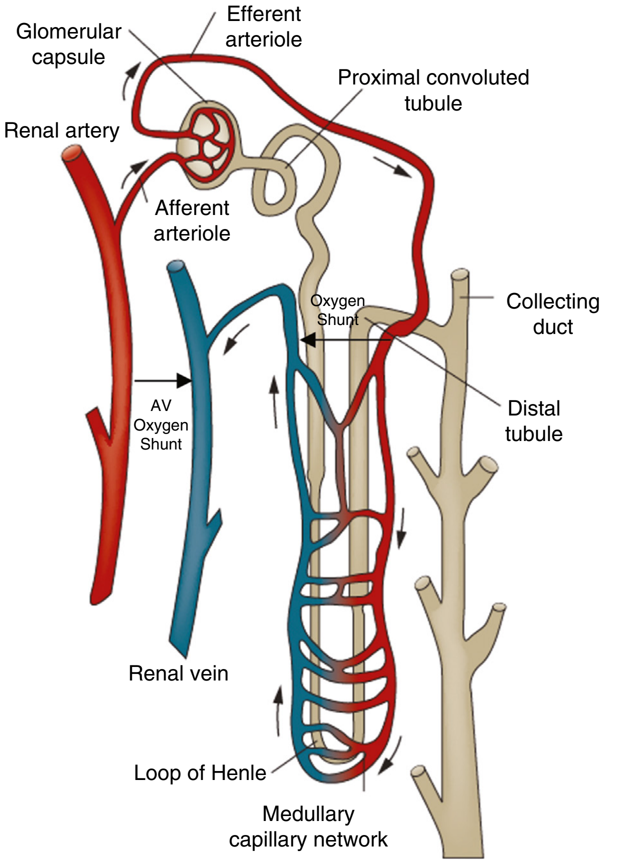

Fig. 3.5 Arterial to venous *(AV)* oxygen shunts. Red blood cells in arterial vessels release oxygen, which diffuses into the interstitium to reach target cells. Blood with lower oxygen tension passes into the venous vessels. The juxtaposition of the arteries and veins present in the kidney facilitates oxygen diffusion from the artery to the vein. Thus oxygen tensions in the capillaries are relatively low when red blood cells reach the peritubular capillary plexus, particularly in the medullary capillary network and interstitium. (From Mimura I, Nangaku M. The suffocating kidney: tubulointerstitial hypoxia in end-stage renal disease. *Nat Rev Nephrol*. 2010;6:667–678.)

arrangement between the descending and ascending vasa recta permits further shunting of oxygen, leaving po_2 values in the medullary tissues of 20 mm Hg or lower toward the papillary tip.[26,28-30] While the physiologically low tissue po2 makes the kidney an ideal oxygen sensor and a major site of body erythropoietin production, it also causes susceptibility to hypoxic damage.[29]

About 75% of the oxygen consumption by the kidneys provides the energy required by the Na^+-K^+-ATPase, which is the major active transport system of the tubules. Changes in oxygen consumption rate vary proportionally with the changes in net Na transport by the tubules. Reduced tissue oxygen levels occur in hypertension, which can compromise renal function.[27,31] The shunting of oxygen from arterioles to venules indicates that diffusible gas molecules that are formed in the kidney, including CO_2, NO, and hydrogen sulfide (H_2S), may also undergo shunting, but from venules to arterioles, thus making it more difficult to wash out unwanted substances such as CO_2 but also accentuating intrarenal retention of protective molecules such as nitric oxide (NO).[32]

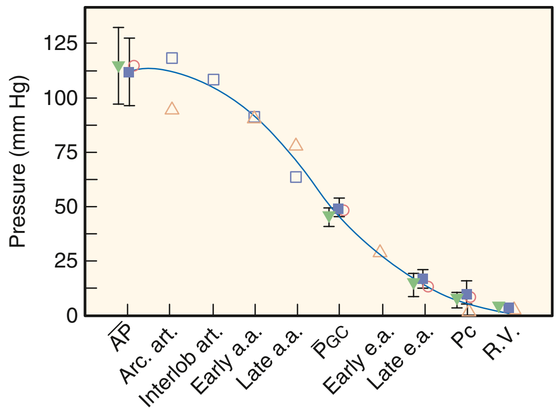

Fig. 3.6 Hydraulic pressure profile along the kidney vasculature. Filled squares and triangles denote values (mean ± 2 standard deviation) obtained from euvolemic and hydropenic Munich-Wistar rats. Values from studies in the squirrel monkey[85] are shown as *open diamonds*. *Open inverted triangles* and *open squares* are from studies in the Sprague-Dawley rat in juxtamedullary nephrons perfused with whole blood. These nephrons are located inside the cortical surface opposed to the pelvic lining and arcuate veins, in which entire pressure profiles can be obtained from the interlobular artery *(Interlob Art),* the proximal *(Early a.a.)* and distal *(Late a.a.)* portions of the afferent arteriole, the glomerular capillaries *(P_{GC})*, the proximal *(Early e.a.)* and late *(Late e.a.)* segments of the efferent arteriole, the peritubular capillaries *(P_c)*, and the renal vein *(R.V.)*. *AP,* Arterial pressure; *Arc. art.,* arcuate artery; *P_{GC}*, glomerular capillary pressure. (Modified from Maddox DA, Brenner BM. Glomerular ultrafiltration. In: *Brenner and Rector's The Kidney.* 7th ed. Philadelphia: W.B. Saunders Company; 2004:353–412; Casellas D, Navar LG. In vitro perfusion of juxtamedullary nephrons in rats. *Am J Physiol.* 1984;246:F349–F358; Imig JD, Roman RJ. Nitric oxide modulates vascular tone in preglomerular arterioles. *Hypertension.* 1992;19[6 Pt 2]:770–774.)

HYDRAULIC PRESSURE PROFILE AND VASCULAR RESISTANCES

The decline in hydraulic pressure along the arterial vasculature to the entry of the afferent arterioles in both the superficial and juxtamedullary regions can be as much as 25 mm Hg at normal perfusion pressures, mostly along the interlobular arteries (Fig. 3.6). However, most of the preglomerular hydraulic pressure drop occurs along the afferent arterioles due to their smaller diameters.[33,34] Approximately 70% of the postglomerular hydraulic pressure drop occurs along the efferent arterioles. The late portion of the afferent arteriole (last 50–150 μm) and the early portion of the efferent arteriole (first 50–150 μm) provide the major fraction of the total preglomerular and postglomerular resistance (see Fig. 3.6).[33,34] Multiphoton imaging studies have indicated the presence of an intraglomerular precapillary sphincter at the terminal end of afferent arterioles (Fig. 3.7).[35,36] Collectively, the total resistance (R_T) consists of two major sites, the afferent (Ra) and efferent (Re) arterioles, and a minor contribution from the outflowing venules and veins (Rv). Accordingly, the following relationships describe the intrarenal cortical vascular resistances:

$$R_T = Ra + Re + Rv \quad [1]$$

$$Ra = \left(AP - P_{GC}\right) / RBF \quad [2]$$

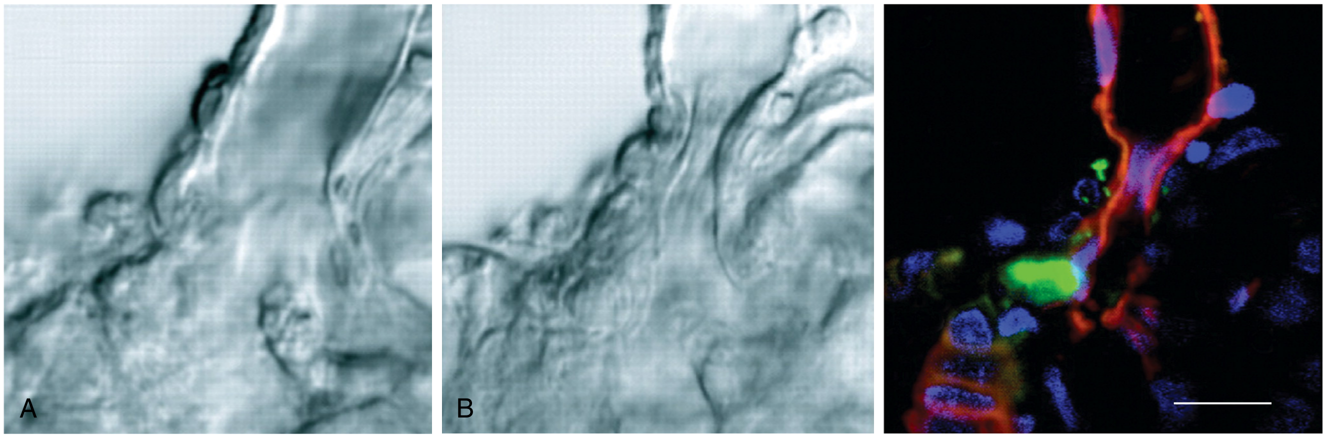

Fig. 3.7 Constriction of the terminal afferent arteriole *(AA)* via an intraglomerular precapillary sphincter in response to elevations in distal tubular NaCl content. (A, B) Transmitted light differential interference contrast (DIC) images. (A) Control, with NaCl concentration at the macula densa at 10 mM. (B) NaCl concentration is increased to 60 mM, resulting in an almost complete closure of the AA. (C) Fluorescence image of the same preparation as shown in B. Vascular endothelium and tubular epithelium are labeled with R18 *(red),* renin granules with quinacrine *(green),* and cell nuclei with Hoechst 33342 *(blue).* Note the presence of renin-positive granular cells in the AA sphincter segment and the highly contractile responses of the terminal portion of the AA. *G,* Glomerulus; *MD,* macula densa. Scale bar = 10 μm. (From Peti-Peterdi J. Multiphoton imaging of renal tissues in vitro. *Am J Physiol Renal Physiol.* 2005;288:F1079–F1083.)

$$Re = \left(P_{GC} - P_C\right) / EABF \quad [3]$$

where $EABF = RBF - GFR$

$$Rv = \left(P_C - P_V\right) / RBF \quad [4]$$

where AP = arterial pressure, P_{GC} = glomerular capillary pressure, P_C = peritubular capillary pressure, P_V = renal vein pressure, and $EABF$ = efferent arteriolar blood flow.

INTRARENAL BLOOD FLOW DISTRIBUTION

The cortex accounts for filtration at the glomeruli and most of the reabsorption from proximal and distal nephron cortical tubules, whereas the medulla reabsorbs less than 20% of the total reabsorbate, in keeping with its primary function to maintain a hypertonic interstitial gradient when needed to excrete concentrated urine. Blood flow to these regions is differentially regulated in response to the differing functions and demands of these two kidney regions.[37] Approximately 80% of the RBF perfuses the cortex and is under the control of numerous intrinsic paracrine vasoactive factors, as well as extrinsic humoral and neural influences. Vasoconstrictors, including angiotensin II (Ang II), endothelin, purinergic agents, and norepinephrine, as well as vasodilators, including bradykinin and nitric oxide, interact to regulate cortical blood flow and medullary blood flow.[37,38] There can be extensive redistribution of blood flow in the kidney under various conditions that may be important in physiologic and pathophysiologic conditions.[39]

Structural differences in vascular components of the cortex and the medulla may account for differences in RBF due to differences in the organization of the afferent and efferent arterioles of the cortical and juxtamedullary glomeruli. The cortical afferent arterioles have larger internal diameters than the efferent arterioles, whereas juxtamedullary afferent and efferent arterioles are larger than the outer cortical arterioles, and efferent arterioles of juxtamedullary nephrons are more muscular compared with the cortical arterioles.[37,40] In addition, the cortical peritubular capillaries, derived from efferent arterioles of cortical glomeruli, are about half the diameter of the medullary vasa recta derived from efferent arterioles of the juxtamedullary glomeruli (Fig. 3.8).[37] These features may partially explain the differential control of medullary and cortical blood flows.

VASCULAR-TUBULE RELATIONS

Cortical vascular-tubule relations have been described extensively (see Figs. 3.1 and 3.2).[20,41,42] The efferent peritubular capillary network and the tubules arising from each glomerulus in the outer regions of the cortex are tightly associated, but this relationship becomes dissociated in deeper regions of the cortex. Although superficial nephron segments and peritubular capillaries arising from the same glomerulus are closely associated, each postglomerular efferent arteriole may serve segments of more than one nephron.[43] However, the loops of Henle of all nephrons, as they descend into the medullary ray, are supplied by postglomerular blood vessels emerging from midcortical and deep nephrons, with branches from deep nephrons descending into the medullary ray termed the *vasa recta* (see Figs. 3.1 and 3.2). The dissociation between individual tubules and the corresponding postglomerular capillary network is most apparent in the inner cortex. The convoluted tubule segments of these nephrons lie above the glomeruli surrounded by the dense network close to the interlobular vessels and by capillary networks arising from other inner cortical glomeruli. With regard to efferent vessel patterns and vascular-tubule relationships,[42,44] there is a close association between the initial portions of peritubular capillaries and proximal tubule segments of the same glomerulus.[45-47]

STRUCTURAL AND FUNCTIONAL ASPECTS OF THE GLOMERULAR MICROCIRCULATION

Some of the structural relationships of the glomerular microcirculation are seen in Fig. 3.8, which shows a

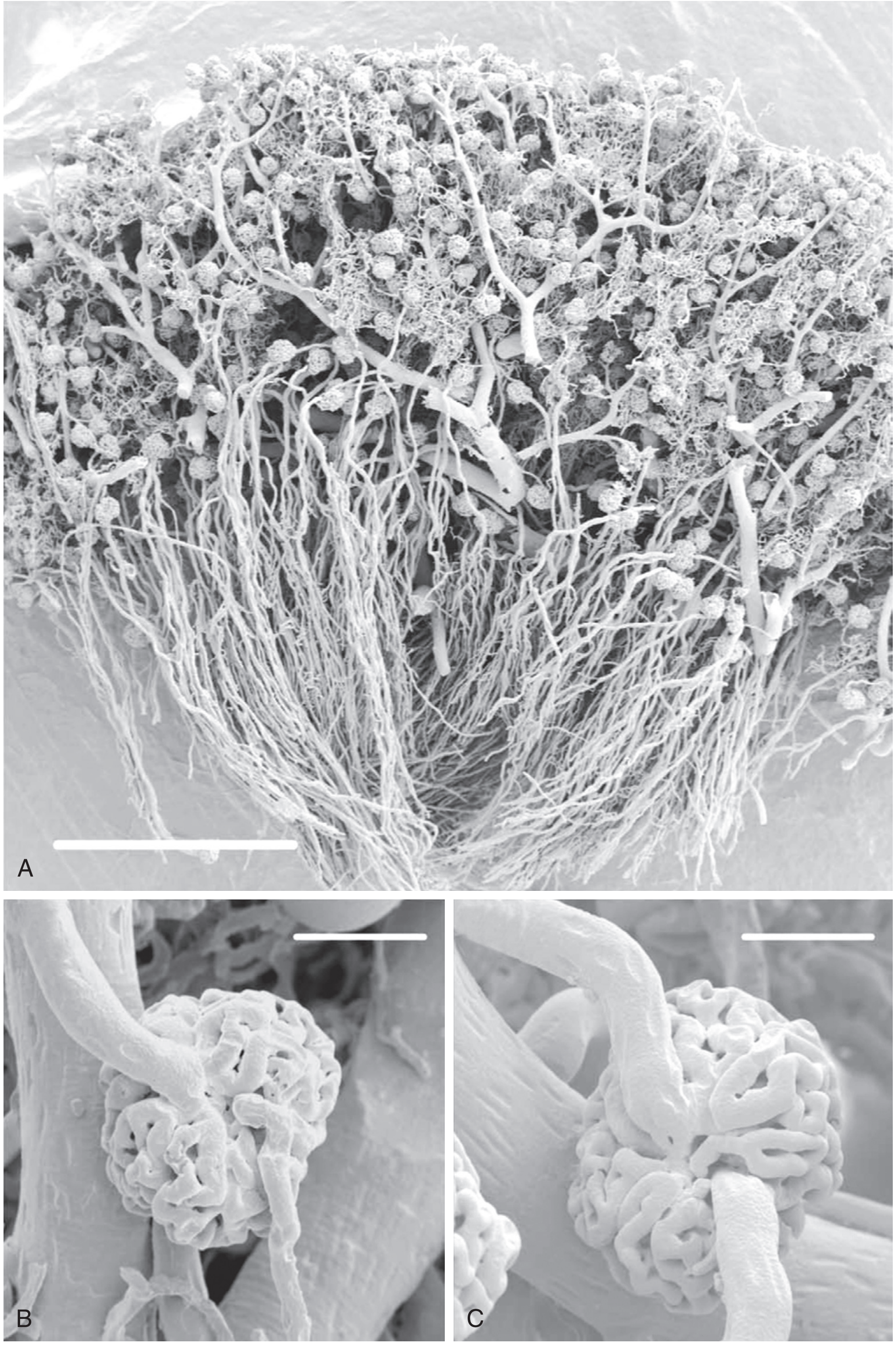

Fig. 3.8 (A) Resin cast of renal glomeruli of a rabbit. (B) Cortical glomerulus showing afferent (upper vessel) and efferent arterioles and the capillary tuft (scale bar = 60 μm). (C) Juxtamedullary glomerulus showing afferent (upper vessel) and efferent arterioles and the capillary tuft (scale bar 60 = μm). Note that the juxtamedullary arterioles are larger in diameter than the cortical glomerular arterioles, particularly the efferent arterioles. (From Evans RG, Eppel GA, Anderson WP, Denton KM. Mechanisms underlying the differential control of blood flow in the renal medulla and cortex. *J Hypertens.* 2004;22:1439–1451.)

scanning electron micrograph of a resin-filled cast of a kidney with afferent arterioles branching from interlobular arteries. The many loops of the glomerular capillaries and the efferent arteriole as it emerges from the glomerular tuft are shown in B and C. [48] Afferent arterioles lose their internal elastic layer and smooth muscle cell layer before entering the glomerular tuft. Smooth muscle cells are replaced by renin-positive, myosin-negative granular cells that are in close contact with the highly contractile cells of the extraglomerular mesangium (Fig. 3.9).[35,49] On entering the Bowman space, afferent arterioles undergo a transition into a vascular chamber that has a manifold-like structure distributing primary branches along the surface of the glomerular tuft, which branch further into the individual glomerular capillaries.[48,50]

The primary branches have wide lumens and immediately acquire features of glomerular capillaries, including a fenestrated endothelium, characteristic glomerular basement membrane, and epithelial foot processes. In human glomeruli, however, these primary branches serve as conduit vessels that branch into filtering capillaries (Fig. 3.10).[50] In contrast, the efferent arteriole originates deep within the tuft, from the convergence of capillaries into multiple lobules that exit into an efferent vascular chamber, which narrows into an efferent arteriole. Additional tributaries join the efferent arteriole as it travels toward the vascular pole. The structure of the capillary wall begins to change, even before the vessels coalesce to form the efferent arteriole, losing fenestrae progressively until a smooth endothelial lining is formed. At the arteriole's terminal portion within the tuft, endothelial cells may bulge into the lumen, reducing its internal diameter.[48] Efferent arterioles acquire a smooth muscle cell layer, which is observed distal to the entry point of the final glomerular capillary. The efferent arteriole is also in close contact with the glomerular mesangium as it forms inside the tuft and with the extraglomerular mesangium as it exits the tuft. This precise and close anatomic relationship between the afferent and efferent arterioles and mesangium with the macula densa cells of the ascending loop of Henle provides the structural basis for the presence of an intraglomerular signaling system, known as the "tubuloglomerular feedback mechanism," which participates in the regulation of blood flow and GFR.[35,48,51]

The appearance of the vascular pathways within the glomerulus may change under different physiologic conditions. The glomerular mesangium (Fig. 3.11) contains contractile elements[48,52] and exhibits contractile activity when exposed to Ang II.[53] Mesangial cells, which possess AT1 receptors for Ang II, undergo contraction when exposed to this peptide in vitro.[54] Three-dimensional reconstruction of the entire mesangium in the rat has suggested that approximately 15% of capillary loops may be entirely enclosed within armlike extensions of mesangial cells that are anchored to the extracellular matrix.[55] Contraction of these cells might alter local blood flow and filtration rate, as well as the intraglomerular

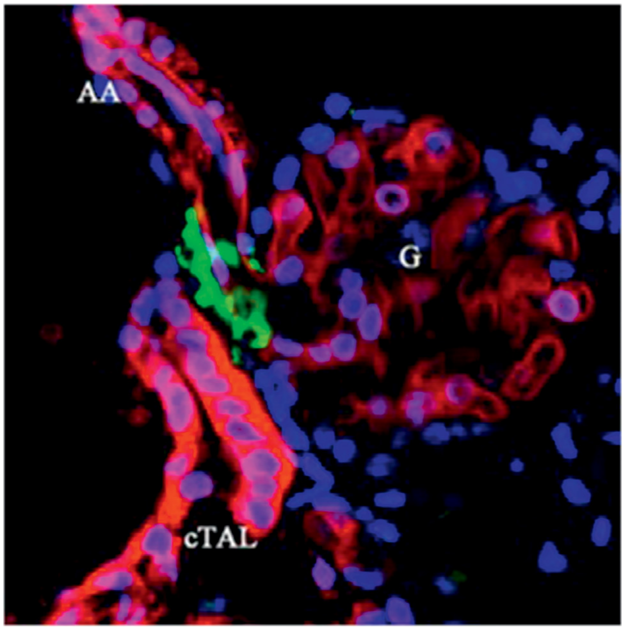

Fig. 3.9 Multicolor labeling of the in vitro microperfused juxtaglomerular apparatus with attached glomerulus. Cell membranes of tubular epithelium (cortical thick ascending limb, *[cTAL]* containing the macula densa), vascular endothelium of the afferent arteriole *(AA),* and glomerulus *(G)* are labeled with R18 *(red),* renin granules with quinacrine *(green),* and cell nuclei with Hoechst 33342 *(blue).* (From Peti-Peterdi J. Multiphoton imaging of renal tissues in vitro. *Am J Physiol Renal Physiol.* 2005;288:F1079–F1083.)

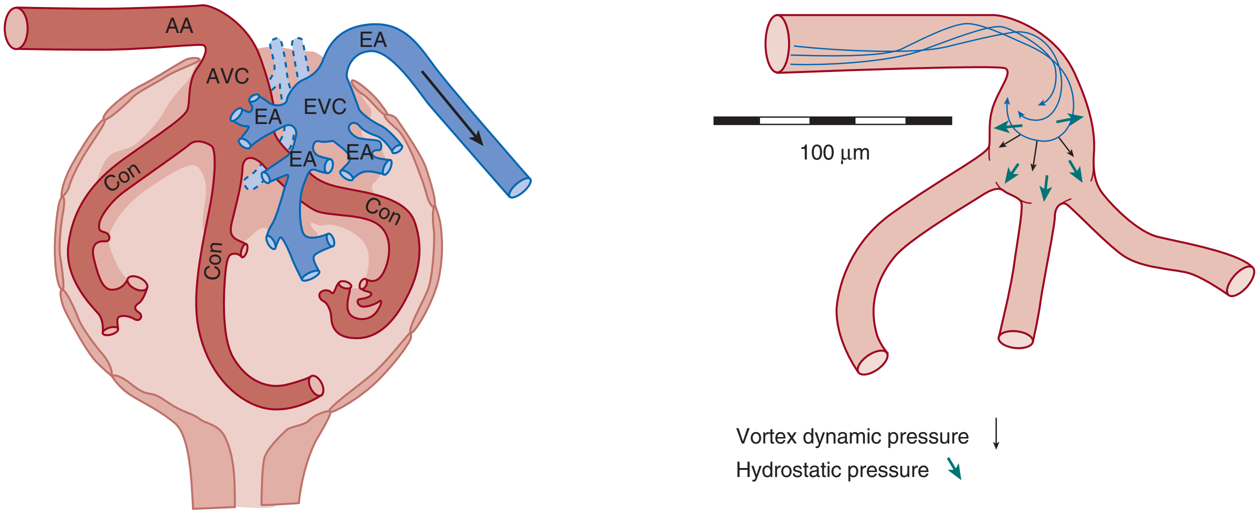

Fig. 3.10 Structure of the human glomerulus. Note the sharp turn made by the afferent arteriole *(AA)* as it enters the afferent vascular chamber *(AVC),* which flows into the conduit afferent capillaries *(Con).* Efferent first-order vessels (E1) and the efferent vascular chamber *(EVC)* transition into the efferent arteriole *(EA). Right,* Distribution of hydraulic forces in the AVC. (From Neal CR, Arkill K, Bell JS, et al. Novel hemodynamic structures in the human glomerulus. *Am J Physiol Renal Physiol.* 2018;315(5):F1370–F1384; color figures courtesy Dr. Christopher Neal.)

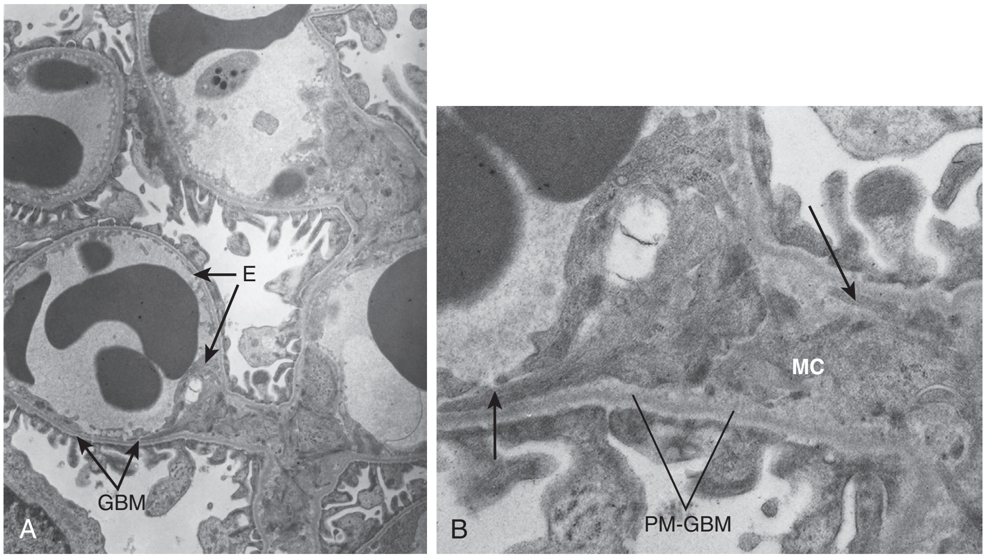

Fig. 3.11 Electron micrographs of glomerular capillaries of a Munich-Wistar rat. (A) An overview of several capillaries (≈×14,500). Most of the glomerular capillary endothelium (E) is in contact with the glomerular basement membrane (GBM) with only a small portion in contact with the mesangium (M). At its outer aspect, podocyte foot processes cover the GBM. There is no basement membrane separating the endothelium from the mesangium at their interface. (B) Mesangial cell (MC) extends outward to meet the glomerular capillary (≈×42,000). These cylinder-like stalks resemble contractile filament bundles *(short arrow)* that attach to the perimesangial glomerular basement membranes (PM-GBM) and extend to the GBM at the mesangial angles *(long arrow)*.[52,113,530] For this preparation, the nephron was perfusion fixed by micropuncture (D.A. Maddox) with 1.25% glutaraldehyde through the Bowman space.

distribution of blood flow and total filtration surface area. Many hormones and other vasoactive substances capable of altering glomerular filtration may bring about this adjustment, in part by altering the state of contraction of mesangial cells.

DETERMINANTS OF GLOMERULAR FILTRATION

A critical function of the mechanisms regulating renal hemodynamics is to maintain the blood flow and pressure profile within the glomerular tufts at levels such that the filtration rate allows optimum function of reabsorptive and secretory processes by the tubular network. The exquisite differential regulation of the preglomerular (afferent) and postglomerular (efferent) resistances exert fine control of the intraglomerular hemodynamic environment and thus the GFR. This nestling of the glomerular capillary system between the afferent and efferent arterioles allows for precise regulation of the intraglomerular forces governing filtration. These forces, coupled with the unique restrictive molecular permeability of the glomerular capillary structure, lead to the formation of a nearly protein-free filtrate, from the glomerular capillaries into Bowman's space, as the first step in the process of urine formation.

PERMEABILITY OF THE GLOMERULAR FILTRATION BARRIER

In addition to the unique role of the glomerular capillary wall and surrounding glomerular epithelial cells in regulating fluid movement from the glomerular capillaries into the Bowman space, these structures also regulate the movement of macromolecules. Molecules the size of inulin or smaller are normally able to cross the glomerular capillary wall without restriction. However, the transglomerular permeation of molecules of increasing size becomes limited, so that molecules the size of albumin or larger are almost completely prevented from crossing into the Bowman space.

The dual critical functions of the glomerular capillary membrane are maintaining a very high hydraulic conductivity to water and small molecules while still maintaining a very low permeability to albumin and other plasma proteins. In order to maintain the albumin stores in the body, less than one-hundredth of one percent (sieving coefficient of .001 or lower) is allowed to permeate the glomerular membrane and pass into the Bowman space under normal conditions. The filtered albumin is then reabsorbed by the megalin/cubulin transport mechanism in proximal tubules. Excess filtration of albumin and other plasma proteins that occurs in various diseases that compromise the integrity of the glomerular membrane overwhelm the protein reabsorption transporters and lead to proteinuria. Although it has been suggested that much greater amounts of albumin are normally filtered, thus presenting a greater load of albumin by the proximal tubule reabsorptive and translocation process,[56] it has been repeatedly pointed out that protein measurements in the filtrate by fluorescence are subject to misinterpretation. All other techniques have consistently shown low and nearly unmeasurable concentrations of albumin in the filtrate.[57,58]

As shown in Fig. 3.12, the composite filtration barrier includes the glycocalyx lining the endothelial cells, the fenestrations of the endothelial layer of the glomerular capillaries, the three layers of the glomerular basement membrane, the filtration slits between adjacent foot processes of the visceral epithelial cells (podocytes) that surround the capillaries, and the filtration slit diaphragm that covers the filtration slits and connects adjacent foot processes to form the ultimate barrier to filtration[59] (see Fig. 3.12). This complex barrier has a high permeability to small molecules such as water, electrolytes, amino acids, glucose, and other

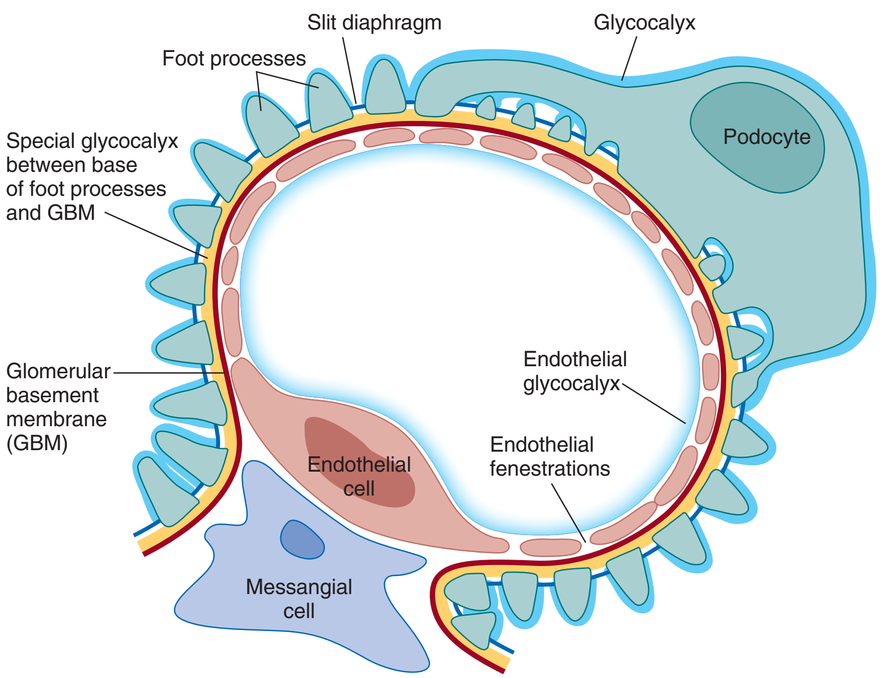

Fig. 3.12 Schematic drawing of a glomerular capillary. The elements considered as part of the glomerular filtration barrier include endothelial glycocalyx, fenestrated endothelium, basement membrane, specialized glycocalyx between the foot processes and basement membrane, and podocyte and slit diaphragm. (From Schlöndorff D, Wyatt CM, Campbell KN. Revisiting the determinants of glomerular filtration barrier: what goes round must come round. *Kidney Int.* 2017;92:533–536.)

endogenous or exogenous compounds with molecular radii smaller than 20 Å. These compounds are freely filtered from the blood into the Bowman space while molecules larger than around 50 Å are restricted.[60-67] In studies using fractional clearances of neutral dextran, comparable size selectivity[68] was observed in rats,[69] dogs,[70] and humans.[71] As emphasized in 2017,[59] there has been considerable controversy regarding the component primarily responsible for the exclusion of macromolecules from the filtrate. The size selectivity of the glomerular filtration barrier is determined largely by a combination of the slit diaphragms between podocyte foot processes and the glomerular basement membrane (GBM). Although there is some controversy regarding glomerular charge selectivity, there is good evidence for the role of charge in restricting the transmural movement of negatively charged macromolecules such as albumin at the level of the GBM, which contains negatively charged heparan sulfate proteoglycans, as well as laminin, type IV collagen, and nidogen.[72,73]

There is also a fine meshwork of glycosaminoglycans covering the luminal endothelial layer and bridging the endothelial fenestrations so that the endothelial layer serves as the initial coarse barrier for macromolecular exclusion.[74] Further studies using negatively charged gold nanoparticles have confirmed that the lamina densa of the basement membrane serves as an exclusion barrier for molecules the size of immunoglobulin G (IgG) and albumin. Smaller particles permeate into the lamina densa and accumulate upstream, covering the base of the foot processes.[66]A labyrinthine fluid space between the underside of podocyte cell body and the foot processes, called the subpodocyte space, may participate in various physiologic (glomerular filtration barrier permeability) and pathologic (glomerulosclerosis) processes.[75]

HYDRAULIC AND ONCOTIC FORCES IN THE GLOMERULAR CAPILLARIES AND BOWMAN SPACE

The process of filtration of fluid at any given point of the glomerular capillary is governed by the net balance among the transcapillary hydraulic pressure gradient (ΔP), the transcapillary colloid osmotic pressure (oncotic pressure) gradient (Δπ), and the hydraulic conductivity of the filtration barrier per surface area unit (Lp), coupled with the surface area (Sf). The product of Lp and Sf is called the "filtration coefficient" (K_f). Fluid flow (J_v) at any given point in the capillary is determined by the Starling equation:

$$J_v = Lp\left[\left(P_{GC} - P_{BS}\right) - \left(\pi_{GC} - \pi_{BS}\right)\right] \quad [5]$$

where P_{GC} and P_{BS} are the hydraulic pressures in the glomerular capillaries and Bowman space, respectively, and π_{GC} and π_{BS} are the corresponding colloid osmotic pressures at any given point. Because the protein concentration of the fluid in the Bowman space is low, π_{BS} approaches zero and can be disregarded. The total GFR (J_v) of fluid for a single nephron (SNGFR) is equal to the product of the surface area for filtration (S_f) and the hydraulic conductivity (Lp), defined as the filtration coefficient (K_f) and the values of the right-hand terms in Eq.5 averaged along the length of the glomerular capillaries yielding the expression:

$$\text{SNGFR} = LpS_f\left[\left(P_{GC} - P_{BS}\right) - \left(\pi_{GC}\right)\right] \textit{ or } \text{SNGFR} = K_f\left(\Delta P - \pi_{GC}\right) \quad [6]$$

Thus

$$\text{SNGFR} = K_f \times P_{UF} \quad [7]$$

where P_{UF} = the mean filtration pressure, the difference between the mean transcapillary hydraulic and colloid osmotic pressure gradients, ΔP and π_{GC}, respectively.

It is important to emphasize the difference between Lp, the hydraulic conductivity, and the macromolecular permeability. These are regulated differently and are not closely coupled. On the basis of known ultrastructural detail and the hydrodynamic properties of the individual components of the filtration barrier, mathematic modeling suggests that only around 2% of the total hydraulic resistance is accounted for by the fenestrated capillary endothelium, whereas the basement membrane accounts for nearly 50%.[72,76,77] The remaining hydraulic resistance resides in the diaphragm of the filtration slits, which are complex structures containing numerous proteins including nephrin and podocin.[72,77-79] Disruption of these slit diaphragm proteins leads to substantial proteinuria.[79] A reduction in the frequency of intact filtration slits is an important factor in the deterioration of filtration in some disease states.[72,80] In the case of macromolecule permeability, mathematic modeling efforts contend that the sieving efficiency of the layers of the glomerular filtration barrier is interdependent. Moreover, whereas hydraulic resistances are additive, macromolecule sieving coefficients are multiplicative; thus a small change in the macromolecule permeability of one layer can significantly change the overall permeability of the filtration barrier.[68,72]

Because surface glomeruli are not present in most experimental animals, indirect approaches to measure glomerular pressure have been used in many experimental studies to evaluate the responses to glomerular pressure. The stop-flow technique has been used by many investigators to estimate P_{GC}.[2,38,81,82] When fluid movement in the early proximal tubule is blocked, intratubular pressure upstream from the block increases until net filtration at the glomerulus ceases.[83] At that point, the sum of this hydrostatic pressure in the early proximal tubule plus the systemic colloid oncotic pressure is equal to the pressure in the glomerular capillaries (P_{GCSF}). The stop-flow technique has been used in different strains of rats, as well as in dogs and mice, with the P_{GCSF} averaging 55 to 60 mm Hg in the dog and about 50 mm Hg in the rat.[4,61] Glomerular capillary pressures calculated using this stop-flow have been compared with values obtained by direct micropuncture of glomerular capillaries in a number of studies; these indicate that P_{GCSF}, measured at normal APs, provides a close estimate of P_{GC} measured directly.[38,61,81,82]

Direct measurements of glomerular capillary hydraulic pressure in vertebrates (P_{GC}) are possible in a mutant strain of the Munich-Wistar rat that has surface glomeruli that allow direct visualization of glomerular capillaries.[84] These studies confirm values for P_{GC} in surface glomeruli of 49 mm Hg in this strain of rats during isovolumia (Fig. 3.13), and similar values were found in the squirrel monkey, which also has some superficial glomeruli.[85] Because the glomerular capillaries are nestled between the afferent and efferent arterioles, P_{GC} is nearly constant throughout the length of the capillary bed, resulting in a transcapillary hydraulic pressure gradient that averages 34 mm Hg in hydropenic Munich-Wistar rats (see Fig. 3.13). Coupling these hydraulic pressure measurements with oncotic pressures determined by the average of afferent and efferent arteriolar protein concentrations allows determination of the hydraulic and oncotic pressures governing filtration along the length of the capillary network.

The early direct measurements of P_{GC} were made in hydropenic rats having surgically induced reductions in plasma volume and GFR indicative of elevated activity of the sympathetic nervous system. Values obtained after plasma volume is restored to a euvolemic state are considered to reflect values in the awake condition[86] and have yielded SNGFR values substantially higher. This is a consequence of increases in P_{GC} and glomerular plasma flow (Q_A) associated with reductions in preglomerular (R_A) and efferent arteriolar (R_E) resistances (see Fig. 3.13).

Glomerular capillary hydraulic and oncotic pressure profiles for rats under hydropenic and euvolemic conditions are shown in Fig. 3.14 using the mean values determined from the studies shown in Fig. 3.13. In hydropenic animals, the plasma oncotic pressure (π_E) rises to a value that equals ΔP such that the local filtration pressure is reduced from approximately 17 mm Hg at the afferent end of the glomerular capillary network to essentially zero by the efferent indicating filtration pressure equilibrium. The value of ΔP is nearly constant along the glomerular capillaries, so the decline in P_{UF} along the capillary network is due to a rise in π_{GC} (see Fig. 3.14). An exact profile of the $\Delta\pi$ curve cannot be ascertained under conditions of filtration pressure equilibrium, and hence only maximum estimates of P_{UF} and minimum estimates of K_f can be obtained by assuming a linear rise in the $\Delta\pi$ curve. In rats expanded with plasma to increase plasma flow and achieve filtration pressure disequilibrium (Fig. 3.14 curve D), it is possible to determine the average P_{UF} and hence K_f.[87] Filtration pressure disequilibrium is the existing condition present in most euvolemic Munich-Wistar rats (see Fig. 3.13 and in most of the studies in Sprague-Dawley rats and in dogs,[4,81,82,88] permitting determinations of P_{UF} along the glomerular capillaries and hence an accurate K_f reflecting the total surface area).

DETERMINATION OF THE FILTRATION COEFFICIENT

As shown in Eq. 7, SNGFR equals the filtration coefficient (K_f) times the net driving force for filtration averaged over the length of the glomerular capillaries (P_{UF}). Values for K_f from studies in euvolemic Munich-Wistar rats are 5.0 ± 0.3 nL/(min·mm Hg), which are similar to those found in other rat strains and in dogs (3–5 nL/[min·mm Hg]).[4,61,81,82] Because this value remains essentially unchanged over a twofold range of changes in Q_A, changes in Q_A, per se, do not affect K_f.[87]

In the rat, total capillary basement membrane surface area per glomerulus (A_s) is around 0.003 cm^2 in superficial nephrons and 0.004 cm^2 in deep nephrons.[89] A large portion of the capillary surface area faces the mesangium and, as a consequence, only the peripheral area of the capillaries surrounded by podocytes participates in filtration. This peripheral area available for filtration (A_p) is only about half that of A_s (~0.0016–0.0018; 0.0019–0.0022 cm^2 in the superficial and deep glomeruli, respectively).[89] Using a value of K_f of around 5 nL/min mm Hg, as determined by micropuncture techniques, with these estimates of A_p, yields a hydraulic conductivity (Lp) of 45 to 48 nL/(sec·mm Hg·cm^2). These estimates of K_f for the rat glomerulus are one or two orders in magnitude higher than those reported for capillary networks in mesentery, skeletal muscle, omentum, or peritubular capillaries of the kidney,[61,90] thus supporting

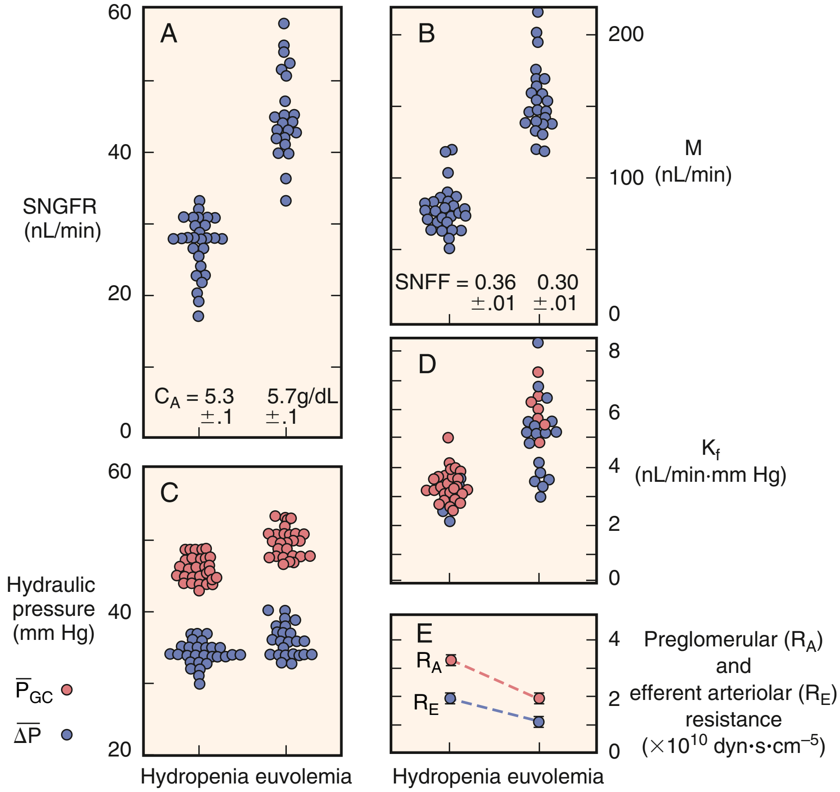

Fig. 3.13 Glomerular filtration in Munich-Wistar rats. (A–E) Each point represents the mean value reported for studies in hydropenic and euvolemic rats provided food and water ad lib until the time of study. Data from euvolemic rats are thought to be representative of nonanesthesia conditions. Only data from studies using male or a mix of male and female rats are shown. Values of the ultrafiltration coefficient, K_f (*red circles* in D), denote minimum values because the animals were in filtration pressure equilibrium. *Blue circles* represent unique values of K_f calculated under conditions of filtration pressure disequilibrium obtained primarily in euvolemic rats ($\pi_E/\Delta P \le 0.95$). *C_A,* Concentration in the afferent arteriole; *ΔP,* pressure gradient; *P_{GC},* pressure in the glomerular capillaries; *Q_A,* glomerular plasma flow rate; *SNFF,* single-nephron filtration fraction; *SNGFR*, single-nephron glomerular filtration. (From Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. *Handbook of Physiology: Section 8; Renal Physiology.* Vol 1. New York: Oxford University Press; 1992:545–638.)

the premise of high glomerular hydraulic permeability of the glomerular capillaries.

DETERMINANTS OF GLOMERULAR FILTRATION COEFFICIENT IN HUMANS

Hydraulic pressure in the glomerular capillaries of human kidneys cannot be measured using micropuncture of glomerular capillaries or measurements of stop-flow pressures in proximal tubules or free-flow pressures in the Bowman space. From determinations of the initial afferent arteriolar oncotic pressure together with the whole-kidney filtration fraction, efferent arteriolar oncotic pressure can be calculated, generally yielding values around 37 mm Hg. In addition, peritubular capillary pressure has been estimated from intrarenal venous pressure measurements, yielding estimates of proximal tubule hydraulic pressure of 20 to 25 mm Hg.[91,92] Coupled with an efferent oncotic pressure of 37 mm Hg, this indicates that the minimal value for glomerular capillary pressure in humans is in the range of 57 to 62 mm Hg. Glomerular volumes and diameters of human kidneys are larger than in the experimental species, suggesting that the single-nephron K_f is greater than in the experimental animals. It is thought that in humans, filtration pressure disequilibrium is normally present, and GFR exhibits less plasma flow dependency than in rats.[2] The molecular sieving approach has been used as an alternative noninvasive means for evaluation of the hydraulic conductivity and filtration coefficient in studies of glomerular dynamics in humans. By using uncharged macromolecules with varying molecular radii, which are partially restricted, the sieving coefficients for molecules of different sizes can be obtained.[69,93,94] Combining the sieving data with mathematic models, estimated values for single-nephron K_f have been derived for normal human subjects and have varied from 3.6 to 9.4 nL/min mm Hg, with average values of about 6 to 7 nL/min mm Hg.[2,4,82,95,96]

SELECTIVE ALTERATIONS IN THE PRIMARY DETERMINANTS OF GLOMERULAR FILTRATION

The four primary determinants of filtration are Q_A, ΔP, K_f, and π_A, and alterations in each of these will affect the GFR. The degree to which such alterations will modify SNGFR has been examined by mathematic modeling[87] and compared with values obtained experimentally (see Arendshorst and Navar,[2] Oken, Navar and colleagues,[4] and Lowenstein and colleagues[91]).

Glomerular Plasma Flow (Q_A)

Because protein is normally excluded from the glomerular filtrate, the total amount of protein entering the glomerular

GLOMERULAR PRESSURES IN THE MUNICH-WISTAR RAT

	Hydropenia		Euvolemia	
	Afferent end	Efferent end	Afferent end	Efferent end
P_{GC}	46	46	50	50
P_{BS}	12	12	14	14
π_{GC}	17	34	19	33
P_{UF}	17 mm Hg	0 mm Hg	17 mm Hg	3 mm Hg

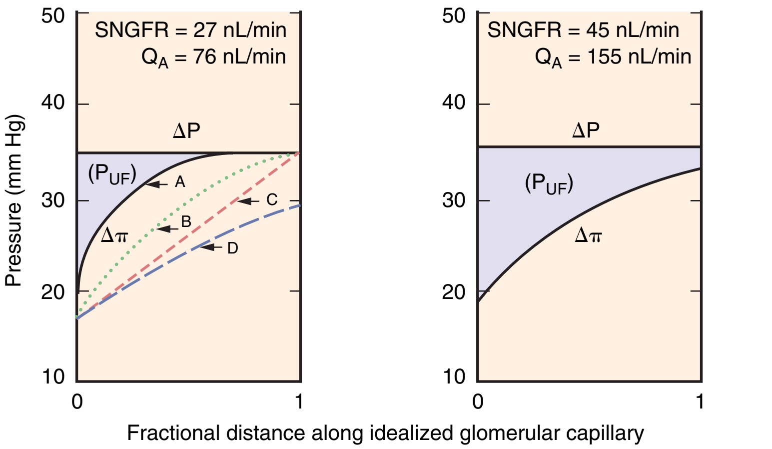

Fig. 3.14 Hydraulic and colloid osmotic pressure profiles along idealized glomerular capillaries in hydropenic and euvolemic rats. Values shown are mean values derived from the studies shown in Fig. 3.13. The transcapillary hydraulic pressure gradient, ΔP, is equal to P_{GC} – PT, and the transcapillary colloid osmotic pressure gradient, $\Delta\pi$, is equal to $\pi_{GC} - \pi_{BS}$, where P_{GC} and P_{BS} are the hydraulic pressures in the glomerular capillary and the Bowman space, respectively, and π_{GC} and π_{BS} are the corresponding colloid osmotic pressures. Because the value of π_{BS} is negligible, $\Delta\pi$ essentially equals π_{GC}. P_{UF} is the ultrafiltration pressure at any point. The area between the ΔP and $\Delta\pi$ curves represents the net ultrafiltration pressure, P_{UF}. *Left,* Lines A and B represent two of the many possible profiles under conditions of filtration pressure equilibrium; line D represents disequilibrium and line C represents the linear $\Delta\pi$ profile. *Right,* Pressure profile after correction for surgical-induced loss of plasma volume depicts a small positive ΔP at the efferent level indicative of disequilibrium conditions. *Q_A,* Glomerular plasma flow; *SNGFR,* single-nephron glomerular filtration rate.

capillary network from the afferent arteriole is maintained, leading to progressively increasing protein concentration as the plasma traverses the glomerular capillaries to the efferent arteriole. The resultant increase in colloid osmotic pressure (πg) counteracts the net hydraulic filtration pressure and may completely offset the ΔP so that $\pi g = \Delta P$ if equilibrium is reached before the plasma reaches the efferent arteriole. Under this condition, where $\Delta P = \pi_E$, SNGFR will vary directly with changes in Q_A as a greater filtering surface area is recruited. Once Q_A increases enough to produce disequilibrium, π_E becomes less than ΔP, and the SNGFR will no longer vary linearly with Q_A.[97] However, there is still a lesser effect of increases in plasma flow to increase GFR, because the filtration fraction will decrease and there will be a lower overall increase in πg. As shown in Fig. 3.15, increases in plasma flow are associated with increases in GFR in many studies in rats, dogs, nonhuman primates, and humans.

Transcapillary Hydraulic Pressure Difference (ΔP)

Mathematic modeling indicates that isolated changes in the glomerular transcapillary hydraulic pressure gradient exert strong effects on SNGFR.[2,4,87] In particular, when ΔP exceeds the colloid osmotic pressure at the efferent end of the glomerular capillary, filtration occurs throughout the glomerular capillary network and SNGFR increases as ΔP increases. The relationship between SNGFR and ΔP is nonlinear, however, because the rise in SNGFR at any given fixed value of Q_A results in a concurrent increase in $\Delta\pi$. Because net effective filtration pressure is a small fraction of P_{GC}, small isolated changes in P_{GC} can cause large percentile changes in net filtration pressure.

Glomerular Capillary Filtration Coefficient (K_f)

Glomerular damage from a variety of kidney diseases and various hormonal and pharmacologic influences can result in alterations in the glomerular filtration coefficient (K_f) due to reductions in surface area available for filtration and/or reductions in the hydraulic conductivity because of thickening of the basement membrane or other derangements. Hydraulic conductivity of the glomerular basement membrane demonstrates an inverse relationship to ΔP, indicating that K_f may be directly influenced by ΔP.[98] K_f is also affected by the plasma protein concentration.[88,99] Under conditions of filtration pressure equilibrium, reductions in K_f do not affect SNGFR until K_f is reduced enough to produce filtration pressure disequilibrium. However, increases in K_f above normal will increase SNGFR until equilibrium conditions occur.[87,90] When plasma flow is high, and disequilibrium exists, there is a direct relationship between K_f and SNGFR.[87,90]

Colloid Osmotic Pressure (π_A)

SNGFR and filtration fraction (SNFF) are each predicted to vary reciprocally as a function of π_A.[87] If Q_A, ΔP, and K_f are held constant, reductions in π_A are predicted to increase P_{UF}, leading to an increase in SNGFR. An increase in π_A should produce a

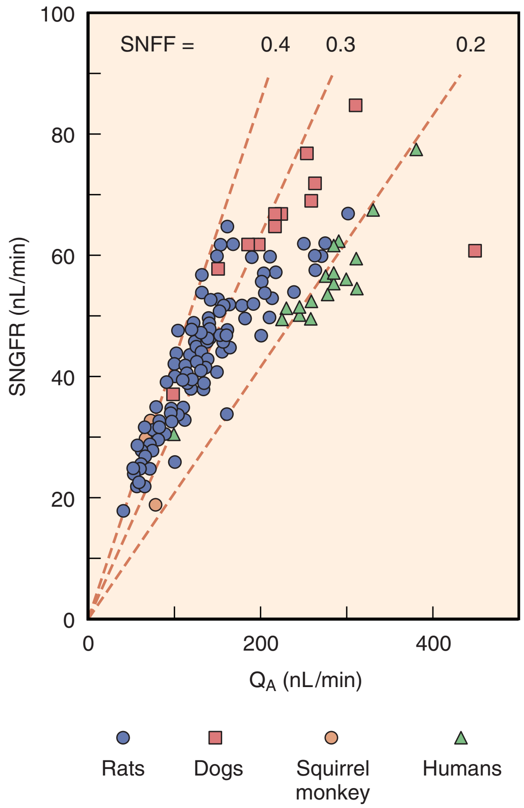

Fig. 3.15 Association between single-nephron glomerular filtration rate *(SNGFR)* and glomerular plasma flow (Q_A); data are from rats, dogs, squirrel monkeys, and humans. The values for SNGFR and Q_A for humans were calculated by dividing the whole-kidney GFR and renal plasma flow by the estimated total number of nephrons/kidney (1 million). Each point represents the mean value for a given study. *SNFF,* Single-nephron filtration fraction. (Data from Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412; Maddox DA, Deen WM, Brenner BM. *Handbook of Physiology: Section 8; Renal Physiology.* Vol 1. New York: Oxford University Press; 1992.)

decrease in SNGFR until π_A equals ΔP (normally, ~35 mm Hg), at which point filtration stops. In contrast to theoretical predictions, experimentally induced reductions in π_A do not increase SNGFR because there are changes in P_{BS} and a decrease in K_f, thereby offsetting the effects of increases in P_{UF} that occur with decreases in π_A.[90] Studies in isolated glomeruli have indicated that extremely low concentrations of albumin produce an increase in K_f, whereas extremely high concentrations of albumin result in a decrease in K_f.[90] However, in vivo studies have shown that increases in plasma π will increase K_f in both rats[99] and dogs.[88] These divergent results of the effects of protein concentration or π_A on K_f can be partially explained by the results from studies of isolated glomerular basement membranes, which have shown a biphasic relationship between albumin concentration and hydraulic permeability.[98] There were lower values of hydraulic permeability at an albumin concentration of 4 g/dL than at either 0 or 8 g/dL.

POSTGLOMERULAR CIRCULATION

PERITUBULAR CAPILLARY DYNAMICS

The Starling forces that control fluid movement across all capillary beds govern the rate of fluid movement across the peritubular capillary walls of the renal cortex. Owing to the high resistance of the afferent and efferent arterioles, a large drop in hydraulic pressure occurs so that peritubular capillary pressure is 15 to 20 mm Hg. Because protein-free fluid is filtered out at the glomerular capillaries, the plasma proteins become concentrated, yielding an increased oncotic pressure of blood flowing into the peritubular capillaries. As a consequence, the balance between the transcapillary oncotic and hydraulic pressure gradients favors movement of the tubular reabsorbate into the capillaries. However, variations in these forces have significant effects on net proximal reabsorption.[38,60,100] The absolute amount of movement resulting from this driving force also depends on the peritubular capillary surface area available for fluid uptake and the hydraulic conductivity of the peritubular capillary wall. Values for the hydraulic conductivity of the peritubular capillaries are not as great as those for the glomerular capillaries, but this difference is offset by the much greater total surface area of the peritubular capillary network.

The peritubular capillary surface contains fenestrations that are bridged by a thin diaphragm and glycocalyx that is negatively charged.[3,101] Beneath the fenestrae of the endothelial cells lies a thin basement membrane that surrounds the capillary. The peritubular capillaries are closely opposed to cortical tubules Fig. 3.16), so the extracellular space between the tubules and capillaries constitutes only about 5% of the cortical volume.[102] The tubular epithelial cells are surrounded by the tubular basement membrane, which is distinct from and wider than the capillary basement membrane. Numerous microfibrils connect the tubular and capillary basement membranes, a feature that may help limit expansion of the interstitium and maintain close contact between tubular epithelial cells and the peritubular capillaries during periods of high fluid flux.[103] Thus the pathway for fluid reabsorption from the tubular lumen to the peritubular capillary is composed, in series, from the epithelial cell, lateral spaces, tubular basement membrane, a narrow interstitial region containing microfibrils, the capillary basement membrane, and the thin membrane bridging the endothelial fenestrae.[103]

Like the endothelial cells, the basement membrane of the peritubular capillaries possesses anionic sites.[101,104] The electronegative charge density of the peritubular capillary basement membrane is significantly greater than that observed in the unfenestrated capillaries of skeletal muscle and similar to that observed in the glomerular capillary bed. These anionic sites in the peritubular capillaries compensate for the greater permeability of fenestrated capillaries, allowing free exchange of water and small molecules while restricting anionic plasma proteins to circulation. The renal peritubular capillaries are reported to be more permeable to both small and large molecules than are other beds,[105] but this may be an artifact of the experimental conditions used. Indeed, other studies have indicated that the permeability of the peritubular vessels to dextrans and albumin is extremely low.[104,106]

MEDULLARY MICROCIRCULATION

Similar to cortical peritubular vessels, the functional role of the medullary peritubular vasculature is to supply the metabolic needs of nearby tissues, but this unique vasculature is also responsible for the uptake and removal of water and

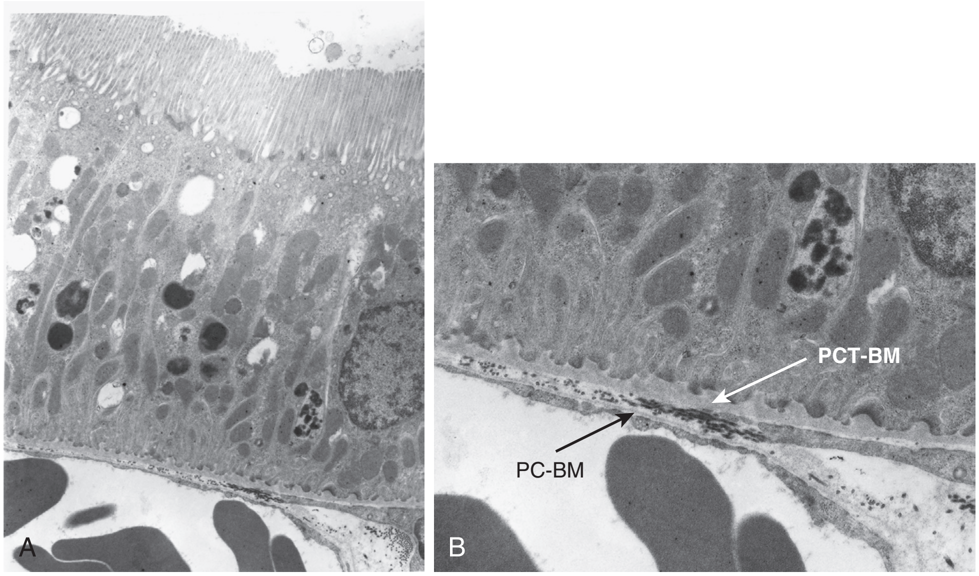

Fig. 3.16 Electron micrographs (by D.A. Maddox) of a proximal tubule of a Munich-Wistar rat. Tubule was perfusion fixed with 1.25% glutaraldehyde, thereby also fixing red cells in adjacent capillaries. (A) The apposition of the basolateral surface of the tubular cells with the adjacent peritubular capillaries is close, leaving little interstitial space where the two come in contact. (≈ ×13,000.) (B) The proximal tubule basement membrane (PC-BM) is relatively thick in comparison with the peritubular capillary endothelial basement membrane (PC-BM.) (≈×25,000.)

solutes extracted from collecting ducts during the process of urine concentration. Because the urinary concentration process requires the development and maintenance of a hypertonic interstitium, the countercurrent arrangement of vasa recta plays a vital role in maintaining the medullary solute gradient through passive countercurrent exchange.

Medullary blood flow constitutes only about 10% to 15% of total RBF[3,6,107,108] and is derived entirely from efferent arterioles of the juxtamedullary nephrons (see Figs. 3.1 and 3.2).[22,40,109-112] Depending on the species and the method of evaluation, from 7% to 18% of glomeruli give rise to efferent arterioles that supply the medulla.[22,113] Efferent arterioles of juxtamedullary nephrons are larger in diameter (see Fig. 3.8), possess thicker endothelium, and have more prominent smooth muscle layers than efferent arterioles originating from superficial glomeruli.[37,114]

Despite the fact that medullary flow is <20% of the cortical flow, it is still relatively high compared with other tissues per gram of tissue; outer medullary flow exceeds that of liver, and inner medullary flow is comparable with that of resting muscle or brain.[115] The high efficiency of countercurrent mechanisms in this area permits the existence and maintenance of the inner medullary solute concentration gradients in the presence of such large flows. The descending vasa recta have a continuous endothelium, in which water moves across water channels, and urea moves through endothelial carriers.[116,117] The ascending vasa recta are fenestrated, with high hydraulic conductivity, and water movement is governed by transcapillary hydraulic and oncotic pressure gradients.[117] Medullary blood flow is highest under conditions of water diuresis and declines during antidiuresis.[107] A direct vasoconstrictive effect of vasopressin on the medullary microcirculation contributes to this decrease during antidiuresis.[118] Vasodilatory factors act to preserve medullary blood flow and prevent ischemia. Acetylcholine,[119] vasodilator prostaglandins,[120] kinins,[121] adenosine,[122] atrial peptides,[123] bradykinin,[9] and nitric oxide[124] increase medullary RBF. In contrast to their vasoconstrictor effects in the renal cortex, Ang II[125-128] and endothelin[125] increase medullary blood flow, effects mediated in part by vasodilatory prostaglandins,[126,127] whereas vasopressin decreases medullary blood flow.[118,129] Alterations in medullary blood flow may be a key determinant of medullary fluid tonicity and, thereby, of solute transport in the loops of Henle and the control of sodium excretion and blood pressure.[130] During hemorrhage, there is primarily cortical ischemia, with maintained blood flow through the medulla.[131]

The precise location of the boundary between the renal cortex and medulla is difficult to discern because the medullary rays of the cortex merge imperceptibly with the medulla. In general, the arcuate arteries or sites at which the interlobular arteries branch into arcuate arteries mark this boundary. The medullary circulation contributes to the countercurrent mechanism by the parallel array of descending and ascending vasa recta. This configuration is characteristic of the inner medulla, but the medulla also contains an outer zone consisting of two morphologically distinct regions, the outer and inner stripes of the outer medulla (see Fig. 3.2). The boundary between the outer medulla and inner medulla is defined by the beginning of the thick ascending limbs of Henle (see Fig. 3.1). In addition to the thick ascending limbs, the outer medulla contains descending straight segments of proximal tubules (pars recta), descending thin limbs, and collecting ducts. The nephron segments of the inner stripe of the outer medulla include thick ascending limbs, thin descending limbs, and collecting ducts. Each of these morphologically distinct medullary regions is supplied and drained by a specific vascular system.

Both the outer and inner stripes contain two distinct circulatory regions—the vascular bundles, formed by the coalescence of the descending and ascending vasa recta, and the interbundle capillary plexus. Vascular bundles of the descending and ascending vasa recta arise from the efferent arterioles of juxtamedullary glomeruli and descend through the outer stripe of the outer medulla to supply the inner stripe of the outer medulla and inner medulla (see Fig. 3.2). Within the outer stripe, nutrient flow is provided by the ascending vasa recta rising from the inner stripe. This notion is supported by the large area of contact between the ascending vasa recta and descending proximal straight tubules within this zone.[110,113,132]

The outer medulla includes the metabolically active thick ascending limbs. Nutrients and O_2 are delivered to this energy-demanding tissue in the inner stripe by a dense capillary plexus arising from a few descending vasa recta at the periphery of the bundles. Of the 10% to 15% of total RBF directed to the medulla, the largest portion perfuses this inner stripe capillary plexus. The descending vasa recta possess a contractile layer composed of smooth muscle cells in the early segments that evolve into pericytes by the more distal portions of the vessels. These pericytes contain smooth muscle α-actin, suggesting that they serve as contractile elements and participate in the regulation of medullary blood flow,[133] as well as in vascular-tubular crosstalk.[134] Each of these vessels also displays a continuous endothelium that persists until the hairpin turn is reached, and the vessels divide to form the medullary capillaries. In contrast, ascending vasa recta, like true capillaries, lack a contractile layer and are characterized by a highly fenestrated endothelium.[135,136] The smooth muscle cells of the descending vasa recta are replaced by pericytes surrounding the endothelium, with subsequent loss of the pericytes and transformation into medullary capillaries accompanied by endothelial fenestrations.[109,132]

The rich capillary network of the inner stripe drains into numerous veins, which, for the most part, do not join the vascular bundles but ascend directly to the outer stripe. These veins subsequently rise to the cortical-medullary junction and join with cortical veins at the level of the inner cortex.[137] A few veins may extend within the medullary rays to regions near the kidney surface.[20,111,137] Thus the capillary network of the inner stripe makes no contact with the vessels draining the inner medulla.

The inner medulla contains the thin descending and thin ascending limbs of Henle, together with collecting ducts (see Fig. 3.2). Within this region, the straight, unbranching vasa recta descend in bundles, with individual vessels leaving at every level to divide into a simple capillary network characterized by elongated links (see Figs. 3.1 and 3.2).[109,113,137] These capillaries converge to form the venous vasa recta. Within the inner medulla, the descending and ascending vascular pathways remain in close apposition, although distinct vascular regions can no longer be clearly discerned. The venous vasa recta rise toward the outer medulla in parallel with the supply vessels to join the vascular bundles. Within the outer stripe of the outer medulla, the vascular bundles spread out and traverse the outer stripe as wide tortuous channels that lie in close apposition to the tubules, eventually emptying into arcuate or deep interlobular veins.[113] The venous pathways in the bundles are both larger and more numerous than the arterial vessels, suggesting lower flow velocities in the ascending (venous) than in the descending (arterial) vessels.[138] The close apposition of the arterial and venous pathways in the vascular bundles is important for maintaining the hypertonicity of the inner medulla.

The mechanism of urine concentration requires coordinated function of the vascular and tubular components of the medulla. In species capable of marked concentrating ability, medullary vascular-tubular relationships show a high degree of organization favoring particular exchange processes by the juxtaposition of specific tubular segments and blood vessels.[137,139] In addition to anatomic proximity, the absolute magnitude of these exchanges is greatly influenced by the permeability characteristics of the structures involved, which may vary significantly among species.[140]

PARACRINE AND ENDOCRINE FACTORS REGULATING RENAL HEMODYNAMICS AND GLOMERULAR FILTRATION RATE

EXTRINSIC AND INTRINSIC REGULATION OF THE RENAL MICROCIRCULATIONS

Various hormonal, neural, and paracrine factors regulate RBF, regional distribution, and GFR.[38,61,90] Blood vessels from the arcuate and interlobular arteries to the afferent and efferent arterioles are influenced to a variable extent by multiple intrinsic and extrinsic factors. As a result, the vascular tones of preglomerular and postglomerular resistance vessels are regulated to control RBF, glomerular hydraulic pressure, and the transcapillary hydraulic pressure gradient. Vasoactive compounds elicit acute alterations in K_f by changing the effective surface area for filtration through contraction of mesangial cells, causing shunting of blood to fewer capillary loops.[38,141,142] In addition, dynamic changes in the actin cytoskeleton of glomerular epithelial cells (podocytes), which contain filamentous actin molecules, may decrease the size of the filtration slit pores, thereby altering hydraulic conductivity of the filtration pathway and reducing K_f.[143] Various growth factors influence chronic changes in renal hemodynamics by promoting mesangial cell proliferation and expansion of the extracellular matrix, leading to obliteration of capillary loops and a reduction in the filtration coefficient.

Our understanding of afferent and efferent arteriolar vascular responses to neural, paracrine, and hormonal vasoactive stimuli have, to a large extent, come from micropuncture studies of glomerular hemodynamics and other methods that examine the effects of vasoactive substances on isolated or exteriorized preglomerular and postglomerular arterioles.[38,61,90,144-151] The results using these different techniques have provided important insights into the vasoactive responsiveness of the microvasculature controlling renal hemodynamics and GFR.

The renal vasculature and glomerular mesangium respond to numerous hormones and vasoactive factors, by altering diameters of the arterioles, and changes in glomerular dynamics or the filtration coefficient. The vasoconstrictors include Ang II, epinephrine, norepinephrine, arachidonic acid metabolites, reactive oxygen species, platelet-activating factor (PAF), adenosine 5'-triphosphate (ATP), endothelin, vasopressin, and serotonin.[38,61,90] Similarly, vasodilatory substances, such as NO and prostaglandin E2 (PGE2), and PGI2, histamine, bradykinin, acetylcholine, insulin, insulin-like growth factor, calcitonin gene-related peptide, cyclic adenosine monophosphate, and relaxin can increase RBF and GFR.[38,61,90] In addition to having direct effects on RBF and GFR, a number of these complex vasoactive systems, such as the renin-angiotensin-aldosterone system (RAAS) and arachidonic acid metabolites, produce both vasoconstriction and vasodilator metabolites and can also stimulate production and release of other factors, thus masking their primary effect. Furthermore, vasoconstrictor stimuli may result in activation of compensatory vasodilatory factors yielding a complex interactive balance regulating renal hemodynamics.

INTRINSIC MECHANISMS: RENAL AUTOREGULATION

Renal autoregulation refers to the intrinsic ability of the kidney to respond to a perturbation that elicits a vasoactive response, which alters renal vascular resistance in the direction that minimizes disruptions in RBF and/or GFR. Changes in perfusion pressure are the manipulation most commonly used to demonstrate autoregulatory behavior. As shown in Fig. 3.17, the kidney autoregulates blood flow over a wide range of perfusion pressures through adjusting afferent arteriolar resistance.

The finding that both RBF and GFR are autoregulated with high efficiency indicates that the principal resistance changes are localized to the preglomerular vasculature. Studies of single-nephron function of superficial nephrons have demonstrated that SNGFR also exhibits efficient autoregulation, as long as the tubular fluid collections do not block flow to the macula densa. Furthermore, direct measurements of glomerular pressures in the Munich-Wistar rat, which has glomeruli on the renal cortical surface that is accessible to micropuncture, have demonstrated autoregulation of glomerular pressure in response to variations in renal arterial perfusion pressure. Fig. 3.18 summarizes the effects of graded reductions in renal perfusion pressure on P_{GC} and preglomerular (R_A) and efferent arteriolar (R_E) resistance.[152] Graded reductions in renal perfusion pressure from 120 to 80 mm Hg resulted in only modest declines in glomerular capillary pressure and blood flow, whereas further reductions in perfusion pressure to 60 mm Hg led to more pronounced declines (see Fig. 3.18).

Autoregulation of glomerular capillary blood flow and P_{GC} is primarily due to the associated changes in R_A, with little or no change in R_E. However, at lower renal perfusion pressures, R_E may contribute to the maintenance of GFR. Under conditions of modest plasma volume expansion, R_A declines while R_E increases slightly as renal perfusion pressure is lowered so that P_{GC} and ΔP are virtually unchanged over the entire range of renal perfusion pressures.[152] The mean glomerular transcapillary hydraulic pressure difference (ΔP) exhibits almost perfect autoregulation over the entire range of perfusion pressures.[152] These observations indicate that autoregulation of GFR is the consequence of the autoregulation of glomerular blood flow and glomerular capillary pressure. Similar results have been obtained in other rat strains and in dogs, where proximal and distal tubular pressures and peritubular capillary pressure also demonstrated autoregulation.[153]

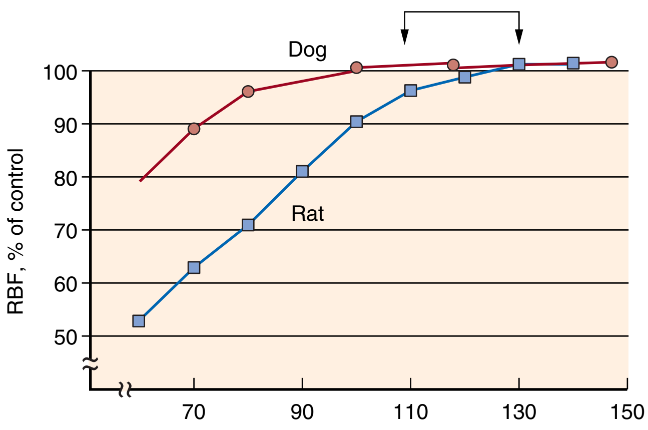

Fig. 3.17 Autoregulatory response of total renal blood flow *(RBF)* to changes in renal perfusion pressure in the dog and rat. In general, the normal anesthetized dog exhibits greater autoregulatory capability to maintain RBF and glomerular filtration rate to lower arterial pressures than the rat. (From Navar LG, Bell PD, Burke TJ. Role of a macula densa feedback mechanism as a mediator of renal autoregulation. *Kidney Int.* 1982;22:S157–S164.)

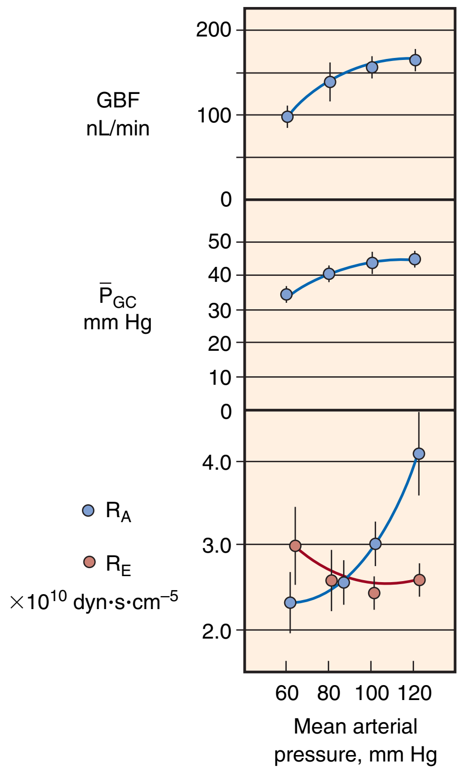

Fig. 3.18 Glomerular dynamics in response to reductions of renal arterial pressure in the normal hydropenic rat. As can be seen, glomerular blood flow *(GBF)* and glomerular capillary hydraulic pressure (P_{GC}) remain relatively constant as blood pressure is lowered from ≈120 to ≈80 mm Hg over the range of perfusion pressure examined, primarily because of reductions in afferent arteriolar resistance *(R_A)*. Efferent arteriolar resistance *(R_E)* was relatively constant but increased slightly at lower pressures. (Modified from Robertson CR, Deen WM, Troy JL, Brenner BM. Dynamics of glomerular ultrafiltration in the rat. III: hemodynamics and autoregulation. *Am J Physiol.* 1972;223:1191, 1972.)

Although more controversial, autoregulation also occurs in the medullary circulation,[154-156] an effect that may be influenced by the volume status of the animal.[155] In the split hydronephrotic rat kidney preparation and in the juxtamedullary nephron preparation,[157,155] reductions in perfusion pressure from 120 to 95 mm Hg elicited dilation of all preglomerular vessels, including the arcuate and interlobular arteries. The large preglomerular arterioles, including interlobular arteries, contribute to the constancy of outer cortical blood flow in the upper autoregulatory range.[158] These responses notwithstanding, most evidence has indicated that the major preglomerular resistance components are the afferent arterioles.[36,159-161] Direct observations of perfused juxtamedullary nephrons revealed parallel reductions in the luminal diameters of arcuate, interlobular, and afferent arterioles in response to elevations in perfusion pressure. However, because quantitatively similar reductions in vessel diameter produce much greater elevations in resistance in smaller than in larger vessels, the predominant effect of these changes is an increase in afferent arteriolar resistance.[33,159]

Under conditions of substantial plasma volume expansion, medullary blood flow autoregulation efficiency is diminished, whereas cortical blood flow autoregulatory responses are maintained. This loss of medullary blood flow autoregulation contributes to the exaggerated pressure natriuresis during plasma volume expansion.[39,162]

Cellular Mechanisms Involved in Renal Autoregulation

Autoregulation of the afferent arteriole and interlobular artery is blocked by administration of L-type calcium channel blockers, inhibition of mechanosensitive cation channels, and a calcium-free perfusate.[163-166] Thus the autoregulatory response involves gating of mechanosensitive channels, which produces membrane depolarization and activation of voltage-dependent calcium channels and leads to an increase in intracellular calcium concentration and vasoconstriction.[163,167,168] Indeed, calcium channel inhibition blocks the autoregulation of RBF.[169,170] Intrinsic metabolites of the cytochrome P450 epoxygenase pathway attenuate the autoregulatory capacity of the afferent arteriole, whereas metabolites of the cytochrome P450 hydroxylase pathway enhance autoregulatory responsiveness.[171]

Inhibition of nitric oxide (NO) does not prevent autoregulation of GFR and RBF, but values for RBF are reduced at any given renal perfusion pressure as compared with control values.[172-175] The transient initial increase in diameter when pressure is increased is of shorter duration during endogenous NO blockade, but the steady state autoregulatory response is unaffected.[168] Cortical and juxtamedullary preglomerular vessels in the split hydronephrotic kidney also autoregulate in the presence of NO inhibition.[176] Thus NO is not essential for the manifestation of renal autoregulation, although it modulates the plateau of the autoregulatory response. NO also plays a role in tubuloglomerular feedback (TGF), as will be discussed.[38,177]

Myogenic and Tubuloglomerular Feedback Mechanisms

Both myogenic and TGF mechanisms contribute to autoregulatory responses. The myogenic mechanism refers to the ability of arterial smooth muscle to contract and relax in response to increases and decreases in vascular wall tension.[160,163,178,179] Thus an increase in perfusion pressure distends the vascular wall and elicits an active contractile response leading to recovery of blood flow toward the control level. While myogenic control of renal vascular resistance contributes substantially to the autoregulatory response, it is not sufficient to completely explain the high-efficiency autoregulation characteristic of renal autoregulation.[160,180]

Autoregulation of RBF is observed, even when TGF is inhibited by furosemide, demonstrating preservation of the myogenic mechanism, which is responsible for the rapid early response in 3 to 10 seconds.[181,182] Autoregulation occurs in all the preglomerular resistance vessels.[159,160,171,182-184] Of note, the afferent arterioles retain the initial contractile response to rapid increases in perfusion pressure when flow to the macula densa is prevented, but they do not exhibit the secondary response.[182] Isolated perfused rabbit afferent arterioles respond to step increases of intraluminal pressure with a decrease in luminal diameter.[150] In contrast, efferent arteriolar segments show either no response or a small increase in diameter, probably reflecting passive physical properties demonstrating that the efferent arteriole does not autoregulate.[163,165,166,176,185] However, efferent arteriolar resistance in vivo may increase in response to prolonged reductions in AP.[152,186] This may result from increased activity of the intrarenal renin-angiotensin system (RAS), which may explain why autoregulation of GFR is more efficient than autoregulation of RBF.

The autoregulatory threshold can be reset in response to a variety of perturbations. Autoregulation is attenuated in diabetic kidneys and may contribute to the hyperfiltration seen early in this disease.[185] Autoregulation is partially restored by insulin treatment and/or by inhibition of endogenous prostaglandin production.[185] Autoregulation in the remnant kidney is markedly attenuated 24 hours after the reduction in renal mass but is restored by cyclooxygenase inhibition, suggesting that release of vasodilatory prostaglandins may be involved in the initial response to increased SNGFR in the remaining nephrons after an acute partial nephrectomy.[187] Much higher pressures than normal are required to evoke a vasoconstrictor response in the afferent arteriole during the development of spontaneous hypertension.[188] Both the afferent arterioles and the interlobular arteries of Dahl salt-sensitive hypertensive rats exhibit reduced myogenic responsiveness to increases in perfusion pressure when fed a high-salt diet.[189] Thus alterations in autoregulatory efficiency occur in a variety of disease states and may influence the kidney's ability to alter excretory responses to increased plasma volume expansion.

The macula densa is a specialized group of about 20 to 30 neuronally differentiated[190] epithelial cells at the end of the thick ascending limb of the loop of Henle on the glomerular side of the tubular wall. It has distinct morphologic characteristics, including a detailed basal cell processes network for cell-to-cell communications[191] and long primary cilia for flow sensing.[192,193] Macula densa cells are adjacent to the cells of the glomerulus and connect with the extraglomerular mesangium and afferent and efferent arterioles of the glomerulus (Fig. 3.19). This anatomic arrangement of macula densa cells, extraglomerular mesangial cells, arteriolar smooth muscle cells, and renin-secreting cells of the afferent arteriole is known as the "juxtaglomerular apparatus" (JGA).[194]

The macula densa is ideally suited to serve as a feedback system whereby a physicochemical stimulus in the tubular fluid activates the macula densa cells, which in turn transmit signals to the arterioles to alter the degree of contraction, thus regulating afferent arteriolar resistance.[195] Changes in the volume flow and composition of the fluid flowing past the macula densa elicit alterations in afferent arteriolar resistance and glomerular filtration, with increases in delivery of fluid resulting in decreases in SNGFR and P_{GC} of the same nephron.[38,196,197] The TGF system senses changes in composition and flow of fluid past the macula densa and "feeds back" signals to afferent arterioles, thus controlling blood flow, glomerular pressure and filtration rate, and providing a powerful feedback mechanism regulating the pressures and flows that govern GFR in response to acute perturbations in delivery of fluid to the macula densa. The TGF mechanism explains the high autoregulation efficiency of RBF and GFR. Increased arterial pressure increases RBF and glomerular capillary pressure, which leads to increased GFR and therefore greater delivery of volume and solute to the distal tubule. Increased distal delivery is sensed by the macula densa, which activates

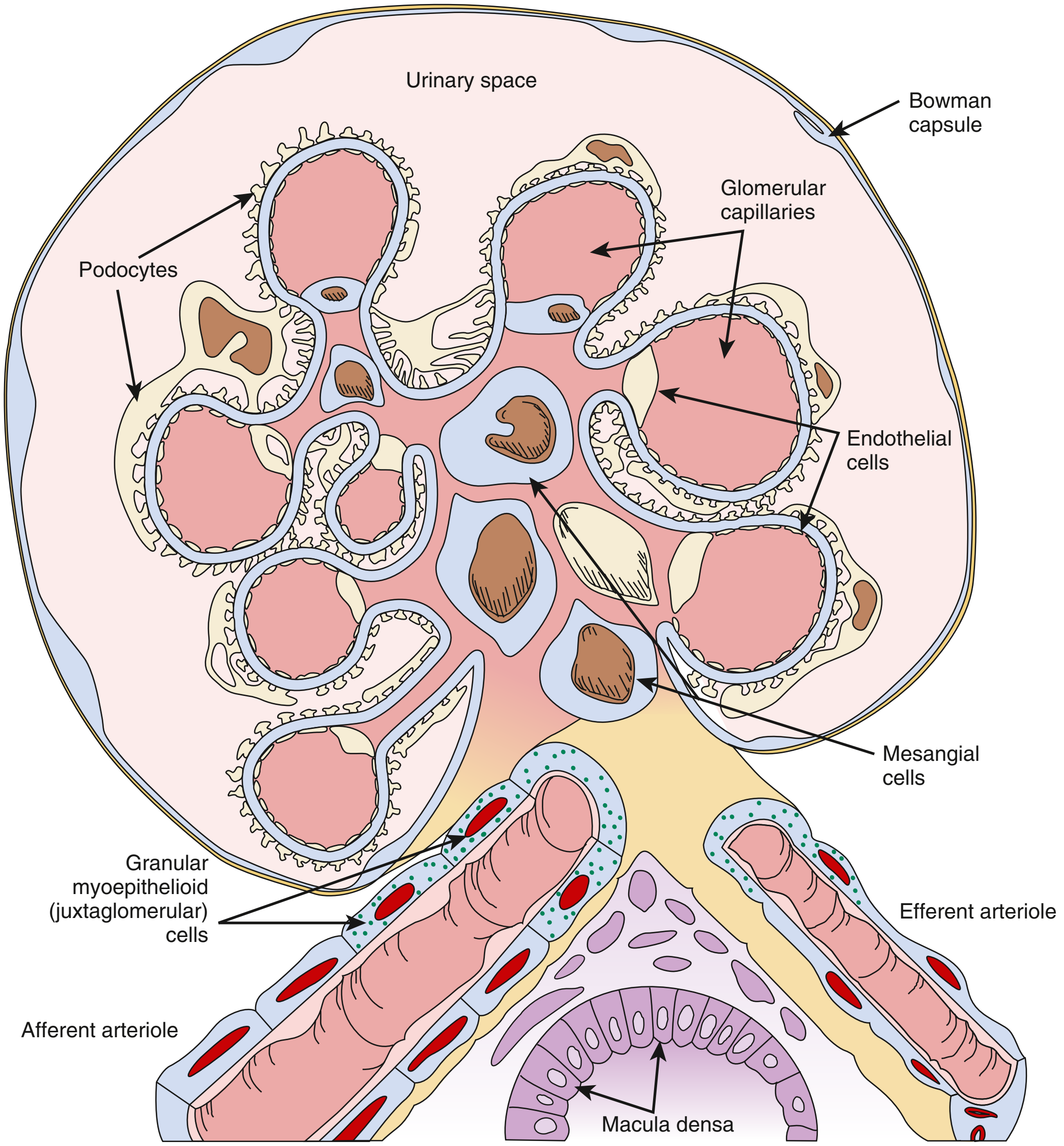

Fig. 3.19 Schematic drawing of a cross-section of a glomerulus, vascular pole, and macula densa cells forming the juxtaglomerular apparatus. As the afferent arteriole enters the glomerular tuft (large vessel, lower left), it breaks into a capillary network, and blood leaves the glomerular tuft via the efferent arteriole (large vessel, lower right). The glomerular capillaries have a fenestrated endothelium. The capillary network and mesangium located between the capillaries are bound together by a common basement membrane (blue line between the podocytes and capillaries). The basement membrane is absent between the capillary lumen and mesangial cells. The outer side of the basement membrane is surrounded by interdigitating visceral epithelial cells known as podocytes. Kriz and coworkers have pointed out that the glomerular mesangium is continuous with the extraglomerular mesangium (consisting of extraglomerular mesangial cells and matrix) at the vascular pole. The extraglomerular mesangium, along with the macula densa cells of the distal tubule and the afferent arteriole, form the juxtaglomerular apparatus. (Courtesy D.A. Maddox.)

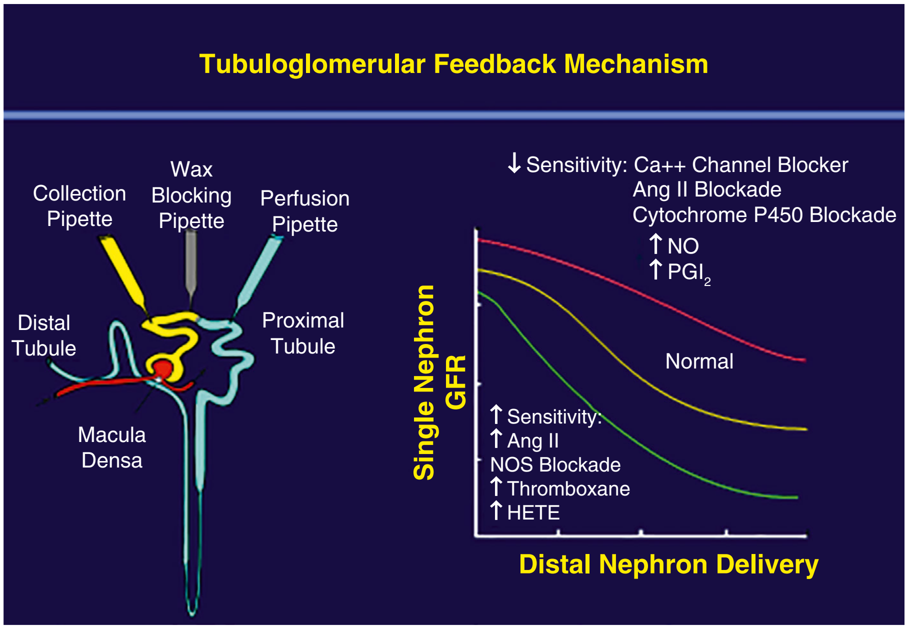

Fig. 3.20 Schematic of representative single-nephron glomerular filtration (SNGFR) rate responses to changes in flow to the distal nephron (TGF) and drawing showing the micropuncture procedure used to change the delivery of fluid to the distal nephron (courtesy Pamela Carmines). The SNGFR decreases with increases in distal volume delivery due to a TGF-mediated increase in afferent arteriolar constriction and subsequent decrease in glomerular pressure with an increase in responsiveness at flows in the normal operating range. TGF responsiveness is dynamic and is modulated by changes in the interstitial environment due to alterations in extracellular fluid volume or level of activity of renin-angiotensin system or other hormonal systems or administration of vasodilator or vasoconstrictor agents. *Ang II,* Angiotensin II; *HETE,* hydroxyicosatetraenoic acid; *NO,* nitric oxide; *NOS,* nitric oxide synthase; *PGI2,* prostaglandin I2.

effector mechanisms that increase preglomerular resistance, reducing RBF, glomerular pressure, and GFR back to optimum levels.[51]

Many studies have demonstrated the roles played by the TGF mechanism (Fig. 3.20). Increased perfusion of the late proximal tubule into the distal tubule causes a reduction in glomerular blood flow, glomerular pressure, and GFR.[198] Experimental maneuvers that decrease distal tubule fluid flow induce afferent arteriolar vasodilation and interfere with the normal autoregulatory response.[38,177,199] In addition, perfusion with furosemide-containing solutions into the macula densa segment abrogates the normal constrictor response of afferent arterioles to increased perfusion pressure,[200] presumably by blocking the Na^+-K^+-$2Cl^-$ transporter on the luminal membrane of the macula densa cells.[160,201] These studies have suggested that the autoregulatory response in juxtamedullary nephrons is also highly dependent on the TGF mechanism. Moreover, deletion of the A_1 adenosine receptor gene in mice to block TGF results in less efficient autoregulation, again indicating the role for TGF in the autoregulatory response.[202]

To examine the role of TGF in autoregulation, investigators have studied spontaneous oscillations in proximal tubule pressure and RBF and the response of the renal circulation to high-frequency oscillations in tubule flow or renal perfusion pressure.[203] Oscillations in tubule pressure have been observed in anesthetized rats at a rate of about three cycles/min that are sensitive to small changes in delivery of fluid to the macula densa.[204] These spontaneous oscillations are eliminated by loop diuretics.[205] To examine this hypothesis,[203] sinusoidal oscillations were induced in distal tubule flow in rats at a frequency similar to that of the spontaneous fluctuations in tubule pressure. Varying distal delivery at this rate caused parallel fluctuations in stop-flow pressure (an index of glomerular capillary pressure), mediated by alterations in afferent resistance, again consistent with dynamic regulation of glomerular blood flow by the TGF system. To investigate the role of this system in autoregulation,[206] the effects of sinusoidal variations in AP at varying frequencies on RBF were examined. Two separate components of autoregulation were identified, one operating at about the same frequency as the spontaneous fluctuations in tubule pressure, the TGF component (kidney specific), and one operating at a much higher frequency consistent with spontaneous fluctuations in vascular smooth muscle tone by the myogenic component (present in all organs, also called vasomotion).[207] These data indicate that slow pressure changes elicit a predominant TGF response, whereas the rapid response reflects the myogenic mechanism.

The TGF mechanism stabilizes delivery of volume and solutes to the distal nephron. Under normal conditions, flow-related changes in the tubular fluid composition at the macula densa are sensed, and signals are transmitted to the afferent arterioles to regulate the filtered load. Early distal tubular fluid is hypotonic (~120 mOsm/kg H_2O), and its composition is closely coupled to fluid flow along the ascending loop of Henle so that increases in flow cause increases in tubular fluid osmolality and NaCl concentration at the macula densa, which lead to vasoconstriction of the afferent arteriole. At the cellular level, increases in tubular fluid osmolality elicit increases in cytosolic [Ca^{2+}] in macula densa cells, which result in release of a vasoconstrictive factor from these cells.[208] As depicted in Fig. 3.21, suggested mediators of TGF include purinergic compounds, such as adenosine and ATP, and one or more of the eicosanoids, such as prostaglandin E2 (PGE2) or 20-hydroxyeicosatetraenoic acid (20-HETE). The factor mediating TGF responses vasoconstricts afferent arteriolar vessels through the opening of voltage-gated Ca^{2+} channels in vascular smooth muscle cells.[38]

The sensitivity of the TGF mechanism can be modulated by many agents and circumstances. TGF sensitivity is diminished during volume expansion, thus allowing greater delivery of fluid and electrolytes to the distal nephron for any given level of GFR. Reductions in TGF sensitivity allow correction of volume expansion. In contrast, contraction of extracellular fluid and blood volume is associated with enhanced sensitivity of the TGF mechanism, which together with augmented proximal reabsorption helps conserve fluid and electrolytes. A major regulator of TGF sensitivity is Ang II. In states of low Ang II activity (e.g., extracellular volume expansion and salt loading), the TGF mechanism is less responsive, whereas feedback sensitivity is enhanced during conditions of high Ang II activity, such as during dehydration, hypotension, or hypovolemia.

The interactions between the myogenic mechanism and TGF are complex and not simply additive. Contributions from other systems add complexity. For example, glomerulotubular balance, whereby proximal tubule

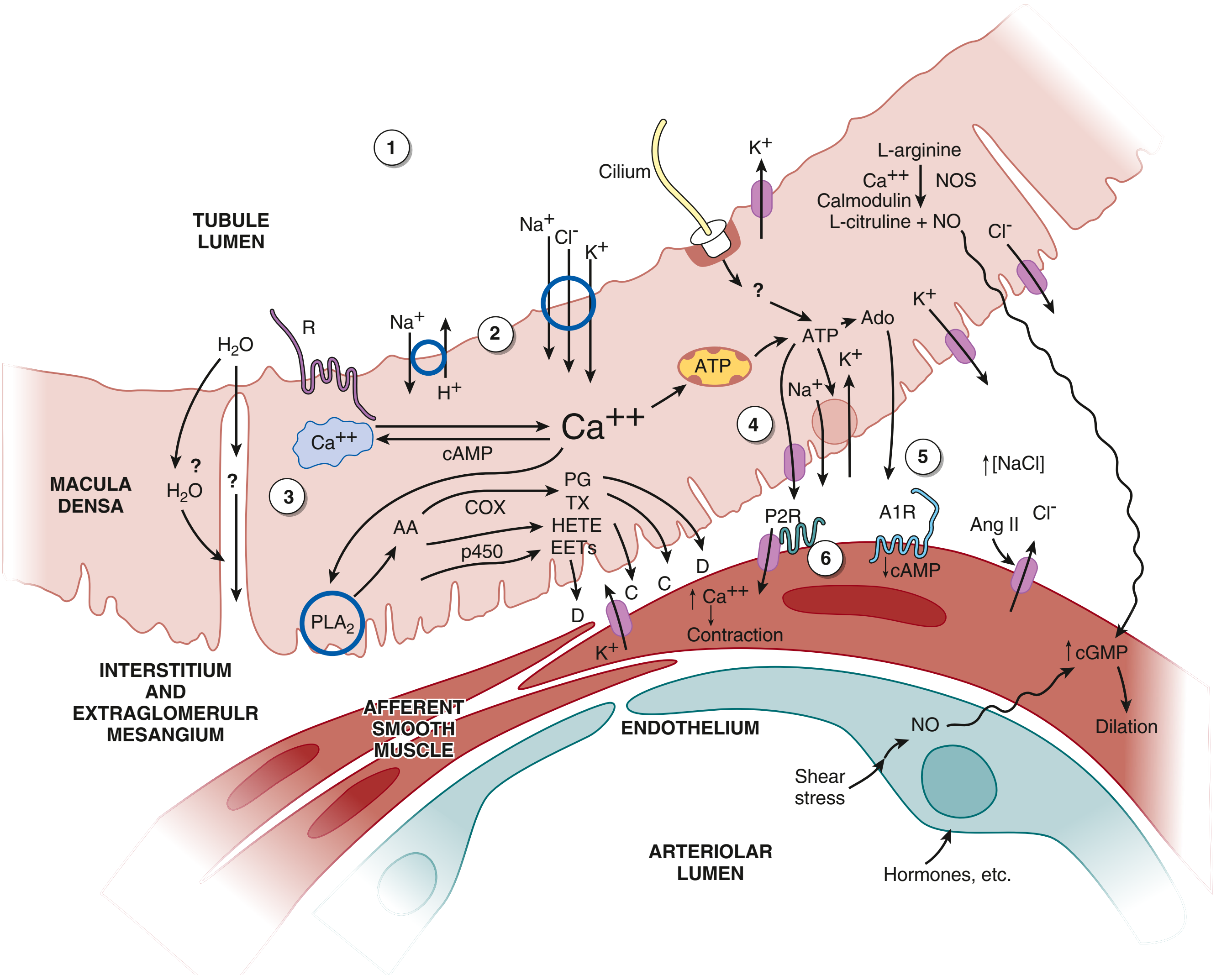

Fig. 3.21 Proposed macula densa tubuloglomerular feedback (TGF) signaling mechanisms. Numbers in circles refer to the following sequence of events: *1,* Flow-dependent changes in tubular fluid composition, including Na^+, Cl^-, osmolality, signals from intratubular paracrine agents, and cilia disturbance; *2,* membrane activation step, including membrane depolarization, enhanced Na^+, Cl^-, K^+ uptake, or other sensing mechanism; *3,* transmission from membrane to intracellular signal mobilization; *4,* formation and release of TGF mediators, including ATP and adenosine *(Ado),* arachidonic acid *(AA)* metabolites, and nitric oxide *(NO)*; *5,* receptor activation by released agents, membrane depolarization, and activation of Ca^{2+} channels in vascular smooth muscle cells; *6,* afferent arteriolar vasoconstriction partially countered by NO-stimulated increases in cGMP and other released mediators—local angiotensin II (Ang II) and neuronal nitric oxide synthase (nNOS) activity modulate the response. *ATP,* Adenosine triphosphate; *A1R,* adenosine A1 receptor; *C,* constriction; *cGMP*, cyclic guanosine monophosphate; *COX,* cyclooxygenase; *D,* dilation; *EET,* epoxyeicosatrienoic acid; *HETE,* hydroxyeicosatetraenoic acid; *P2R,* P2 purinergic receptors; *p450,* cytochrome P 450; *PG,* prostaglandins; *PLA2,* phospholipase A2; *TX,* thromboxanes. As described in the text, the sodium/glucose cotransporter is also present on the luminal membrane. (Expanded from Navar LG, Bell PD, Burke TJ. Role of a macula densa feedback mechanism as a mediator of renal autoregulation. *Kidney Int.* 1982;22:S157–S164.)

reabsorption rate increases as GFR rises, blunts the effects of alterations in GFR on distal delivery. Because the myogenic and TGF responses share the same effector site, the afferent arteriole, interactions between these two systems are unavoidable, and each response is capable of modulating the other.[209] The prevailing view is that these two mechanisms act in concert to accomplish the same end, namely a stabilization of renal function when perturbations in filtered load occur.[38,180,210] Furthermore, the time constraints are different; the myogenic component of autoregulation requires less than 10 seconds for completion and normally follows first-order kinetics without rate-sensitive components.[180] The response time for TGF is slower and may take 30 to 60 seconds, and there are spontaneous oscillations at 0.025 to 0.033 Hz.[180] The myogenic and TGF mechanisms account for most of the autoregulatory responses, but a third system has been proposed.[180] Furthermore, the nature of their interaction may be complex, with the TGF primarily influencing the sensitivity of the myogenic mechanism.[38] In addition to contractile cells, renin cells in the JGA are endowed with a pressure-sensing (baroreceptor) mechanism, a nuclear mechanotransducer that directly couples perfusion pressure and renin gene expression to maintain blood pressure and fluid volume.[211]

Tubuloglomerular Feedback Signals Controlling Renal Blood Flow and Glomerular Filtration Rate

Several factors have been identified as tubular signals for TGF.[212] Changes in delivery of Na^+, Cl^-, and K^+ are thought to be sensed by the macula densa through the Na^+-K^+-$2Cl^-$ cotransporter on the luminal cell membrane of the macula densa cells.[213] Alterations in Na^+, K^+, and Cl^- reabsorption result in inverse changes in SNGFR and renal vascular resistance, primarily due to changes in preglomerular resistance. For example, when salt concentration increases at the macula densa, the feedback mechanism increases afferent arteriolar resistance, thus decreasing glomerular pressure and SNGFR. Agents such as furosemide that interfere with the Na^+-K^+-$2Cl^-$ cotransporter in the macula densa cells[201] inhibit the feedback response.[214]Additional studies in which macula densa segments were perfused with solutions that contained minimal concentrations of essential ions needed to maintain the integrity of the Na^+-K^+-$2Cl^-$ cotransporter, with the remaining solute being either deficient in Na^+ (choline chloride) or deficient in Cl^- (Na isocyanate). These solutions elicited normal TGF responses.[215] Furthermore, orthograde perfusion with nonelectrolyte solutes also elicited TGF responses.[38,216] Collectively, these results indicate that the integrity of the Na^+-K^+-$2Cl^-$ cotransporter must be maintained for the sensing mechanism to function normally. However, the actual sensing mechanism may be activated by changes in total solute concentration. Furthermore, stimulation of the primary cilium in macula densa cells responds to flow-dependent signals and alter the magnitude of the TGF response.[192,193]

The intracellular signaling cascade that generates a vasoactive agent that is secreted remains unclear. Some studies suggest that the luminal signal activates release of Ca^{2+} from the intracellular stores that leads to the formation of ATP, which is secreted and activates purinergic receptors that elicit vascular smooth muscle contraction. This is illustrated in Fig. 3.21, which is derived from several sources.[38,197] According to this scheme, ATP release at the basolateral cell membrane through ATP-permeable, large-conductance anion channels provides the link between macula densa cells and adjacent vascular smooth muscle cells, and is also metabolized further, with ultimate degradation to the metabolites, adenosine diphosphate (ADP), and adenosine monophosphate (AMP). Activity of cytosolic 5′-nucleotidase or endo-5′-nucleotidase bound to the cell membrane results in the formation of adenosine which may also participate in TGF signaling.[197] In addition to the ATP metabolites, the macula densa cells also produce arachidonic acid metabolites, including PGE2 and PGI2, and nitric oxide and reactive oxygen species. Thus several vasoactive substances are secreted and may alter afferent arteriolar vascular tone. Although ATP and adenosine are formed in macula densa cells or in the adjacent interstitium, ATP interacts with purinergic (P2) receptors on the extraglomerular mesangial and vascular cells, resulting in an increase in $[Ca^{2+}]_i$.[217] The increase in $[Ca^{2+}]_i$ may occur, in part, via basolateral membrane depolarization through receptor-operated channels, followed by a further increase in Ca^{2+} entry into the cells via voltage-gated Ca^{2+} channels.[218] ATP can exert direct effects on vascular smooth muscle cells or indirect effects via activation of mesangial cells. Some of the ATP is metabolized to adenosine, which also can directly constrict the afferent arteriole through activation of purinergic P1 receptors.[219]

Although there is general consensus that ATP is secreted by macula densa cells, some studies indicate that the ATP metabolite, adenosine, mediates TGF. Intraluminal administration of an adenosine A1 receptor agonist enhances the TGF response.[220] In addition, TGF is attenuated in adenosine A1 receptor–deficient mice.[213,221,222] However, adenosine also activates adenosine A2 receptors, which cause afferent arteriolar dilation and abrogate the actions of A1 receptors.[223,224]

Efferent arterioles also respond to adenosine, but they vasodilate in response to adenosine via A2 receptors, which antagonize the effects of A1 receptors.[224,225]

Another important paracrine regulator is NO, which exerts vasodilatory responses when released by the macula densa cells. Under normal circumstances, when the NaCl concentration of tubular fluid is increased, there are increases in ATP release coupled with reductions in PGE2 formation until the ATP release reaches a plateau and the PGE_2 release is markedly reduced. With further increases in [NaCl], NO release is augmented to counteract the effects of increased ATP.[38]

In addition to the paracrine factors released by macula densa cells, there are also many modulatory agents that influence the sensitivity of TGF responses. Ang II is one of the more important factors. TGF is blunted by Ang II antagonists and Ang II synthesis inhibitors, and TGF is markedly reduced in knockout mice lacking the AT1A Ang II receptors or angiotensin-converting enzyme (ACE).[226,227] Furthermore, systemic infusion of Ang II in ACE knockout mice restores TGF.[226,228-233] Ang II also enhances TGF via activation of AT1 receptors on the luminal membrane of the macula densa.[234] Acute inhibition of the AT1 receptor in normal mice reduces

TGF responses and reduces autoregulatory efficiency.[229] Adenosine A_1 receptor antagonist administration results in decreased afferent arteriolar resistance and increased transcapillary hydraulic pressure differences (ΔP), whereas pretreatment with an angiotensin AT1 receptor antagonist prevents these changes.[235] While Ang II is not the mediator of TGF, Ang II plays a prominent role in modulating TGF sensitivity mediated through the AT1 receptor.

Neuronal NO synthase (nNOS or NOS1) is present in macula densa cells.[236] NO derived from nNOS in the macula densa provides a vasodilatory influence on TGF, decreasing the vasoconstriction of the afferent arteriole that otherwise would occur.[236,237] Increased distal sodium chloride delivery to the macula densa stimulates nNOS activity and also increases activity of the inducible form of cyclooxygenase (COX-2), which forms PGE2 and counteracts TGF-mediated constriction of the afferent arteriole.[236,237] Macula densa cell pH increases in response to increased luminal sodium concentration and may be related to the stimulation of nNOS.[238] Inhibition of macula densa guanylate cyclase increases the TGF response to high luminal [NaCl], further indicating the importance of NO in modulating TGF.[239] Microperfusion of the macula densa segment with an inhibitor of NO production constricts the adjacent afferent arteriole.[240] Microperfusion of the macula densa with the precursor of NO, L-arginine, blunts TGF responses, especially in salt-depleted animals.[241-243] Thus NO released from macula densa cells or endothelial cells causes afferent arteriolar vasodilation acutely or may blunt TGF responses. An increase in NO production may also inhibit renin release by increasing cyclic guanosine monophosphate (cGMP) in the granular cells of the afferent arteriole,[244] thereby accentuating its vasodilatory effects. When NO production is chronically blocked in knockout mice lacking nNOS, TGF in response to acute perturbations in distal sodium delivery remains normal.[230] However, the presence of intact nNOS in the JGA is required for sodium chloride–dependent renin secretion.[230] The TGF system, which elicits vasoconstriction and a reduction in SNGFR in response to acute increases in sodium and solute delivery to the macula densa, appears to activate a vasodilatory response secondarily via NO release.[245] Interestingly, the sodium-glucose cotransporter SGLT1 is expressed at the macula densa luminal membrane and may function to couple salt and metabolic sensing in conditions of hyperglycemia via NOS1-dependent NO production. An attenuated TGF via the macula densa SGLT1-NOS1 pathway may play a crucial role in the development of glomerular hyperfiltration in diabetes.[246] Two novel treatments for diabetic kidney disease, SGLT2 inhibitors and glucagon-like peptide-1 (GLP-1) receptor agonists, have major effects on renal hemodynamics that contribute to their renoprotective effects (see further discussion on the roles of these compounds in diabetes and kidney disease management in Chapters 40, 49, and 53).[247]

The dynamics of the TGF mechanism can be temporally divided into two or more responses with different time constants. The initial rapid response occurs within a few seconds and elicits rapid vasoconstriction and decrease in GFR and P_{GC} when sodium delivery to the macula densa cells is acutely increased. A second vasoconstrictor response occurs in seconds to minutes and changes the slope of the response to a slower time constant. This may be due to modulation of the initial response by some of the modulating agents mentioned. The rapid TGF system prevents large changes in GFR under conditions such as spontaneous fluctuations in blood pressure, thereby maintaining tight control of distal sodium delivery in the short term. Over the long term, renin secretion, controlled by the JGA in accordance with the requirements for sodium balance and the TGF system, resets to a new sodium delivery rate.[230]

Connecting Tubule Glomerular Feedback Mechanism

There is also growing interest in a second feedback loop that links the connecting tubule, which has been shown to be in apposition to its own glomerulus and in close contact with the afferent arteriole.[248] In vitro perfusion of the connecting tubule with increased luminal NaCl elicits vasodilation of preconstricted afferent arterioles.[249] This action is opposite to the effect of macula densa TGF signaling, raising the question of how these two opposing systems interact. Additionally, amiloride prevents the action of increased luminal Na^+, indicating that the epithelial sodium channel mediates the connecting TGF (CTGF).[249] The afferent vasodilator effect of increased $E_{Na}C$ activity is mediated by PGE2 acting on an EP4 receptor on the afferent arteriole.[250] An additional role of epoxyeicosatrienoic acid (EET) has also been suggested.[251] Furthermore, Ang II in the luminal fluid enhances the afferent arteriolar vasodilator effect caused by increases in luminal Na^+ concentration.[252] This CTGF mechanism mediates resetting of the macula densa TGF mechanism by reducing its sensitivity.[253] A modulating role of CTGF has been observed in experimental hypertension,[254] during high salt intake,[255] and in the renal vasodilatory response that occurs in the remaining kidney after unilateral nephrectomy.[256]

ENDOTHELIAL FACTORS AND GASEOUS TRANSMITTERS CONTROLLING RENAL HEMODYNAMICS AND GLOMERULAR FILTRATION RATE

A growing area of research involves paracrine interactions between endothelial cells and the underlying smooth muscle cells. As shown in Fig. 3.22, endothelial cells respond to various physical and chemical stimuli, including pressure, flow, shear stress, and circumferential strain, as well as vasoactive factors in the blood. Under normal conditions, an increase in shear stress activates endothelial cells to produce NO and the prostanoids PGE2nd PGI2, which help adjust vascular tone to accommodate the increased load. However, during conditions of tissue injury or inflammation, the endothelial cells may also be stimulated to produce endothelin, thromboxane, certain growth and profibrotic factors that elicit vasoconstriction, and/or additional paracrine factors associated with tissue injury and fibrosis. Several of these factors associated with regulation of the renal microcirculation are gaseous physiologic transmitters, called “gasotransmitters,” which have been identified over the last 2 decades. NO is the first such gasotransmitter discovered, but carbon monoxide (CO) and hydrogen sulfide (H_2S) have also been shown to influence renal microcirculation.

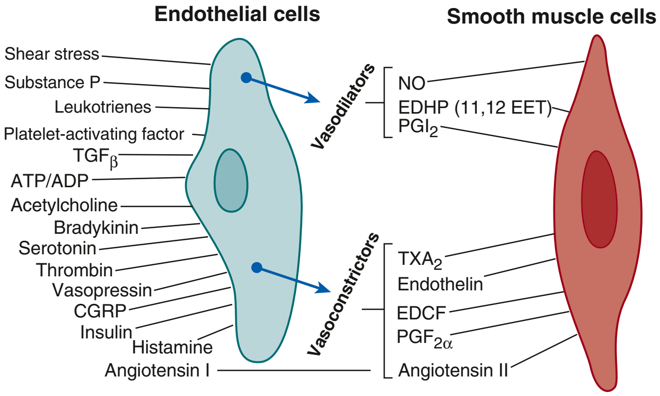

Fig. 3.22 Interaction of endothelial cells with smooth muscle or mesangial cells. As indicated in the text, several agents interact with endothelial cells to produce the vasodilator nitric oxide *(NO)*. Others yield vasoconstriction (see text and other sources). Angiotensin-converting enzyme converts angiotensin I to the potent vasoconstrictor angiotensin II. *ATP–ADP,* Adenosine trisphosphate–adenosine diphosphate; *CGRP,* calcitonin gene-related peptide; *EDCF,* endothelial-derived constricting factor; *EDHP,* endothelial-derived hyperpolarizing factor; *EET,* epoxyeicosatrienoic acid; *$PGF_{2\alpha}$,* prostaglandin $F_{2\alpha}$; *PGI_2,* prostaglandin I2; *TGF_β,* transforming growth factor β; *TXA_2,* thromboxane A2. (Modified from Arendshorst W, Navar LG. Renal circulation and glomerular hemodynamics. In: Schrier RW, Coffman TM, Falk RJ, et al, eds. *Schrier's Diseases of the Kidney*. Philadelphia: Lippincott Williams & Wilkins; 2013:74–131; Navar LG, Arendshorst WJ, Pallone TL, et al. The renal microcirculation. In: Tuma RF, Duran WN, Ley K, eds. *Handbook of Physiology: Microcirculation.* Vol 2. Cambridge, MA: Academic Press; 2008:550–683; Maddox, DA, Deen WL, Brenner BM. *Glomerular Filtration; Handbook of Physiology: Section 8 Renal Physiology*. New York, NY: Oxford University Press; 1992:545–638; Maddox DA, Brenner BM. Glomerular ultrafiltration. In: Brenner BM, ed. *The Kidney*. 7th ed. Philadelphia: Saunders; 2004:353–412.)

Nitric Oxide and Nitric Oxide Synthases

In 1980, Furchgott and Zawadzki[257] demonstrated that the vasodilatory action of acetylcholine requires the presence of an intact endothelium. Acetylcholine binds to receptors on endothelial cells, leading to the formation and release of an "endothelial-derived relaxing factor," now known to be NO.[258,259] Many cell types, including the endothelium, produce NO from the amino acid L-arginine[38,90,260] by a family of nitric oxide synthases (NOSs) that are present in many cell types, including vascular endothelial cells, macrophages, neurons, glomerular mesangial cells, macula densa, and renal tubular cells.[38,261-263] Three main NOS isoforms have been isolated. Neuronal NOS, also termed "NOS1" or "nNOS," and endothelial NOS, also called "NOS3" or "eNOS," are constitutively present in the kidney. A third NOS, iNOS or NOS2, is inducible and expressed after transcriptional induction and remains active for prolonged periods.[38,90] All three isoforms of NOS are found in the kidney. The arcuate and interlobular arteries, as well as the afferent and efferent arterioles, all produce NO, which regulates basal vascular tone, as indicated by the constriction that occurs in response to inhibition of endogenous NO production.[38,90]

Once released by the endothelium, NO diffuses into adjacent and downstream vascular smooth muscle cells,[264] where it stimulates the activity of soluble guanylate cyclase and increases cGMP formation.[90,265-270] cGMP reduces calcium influx, intracellular calcium release, and intracellular calcium concentration. This occurs, in part, through a cGMP-dependent protein kinase (PKG)-mediated phosphorylation of targets, which include inositol trisphosphate (IP3) receptors, calcium channels, and phospholipase A2,[271] thereby reducing the amount of free calcium available for contraction, hence promoting relaxation.[272]

In addition to stimulation by acetylcholine, NO formation in the vascular endothelium increases in response to bradykinin,[269,273-276] thrombin,[277] PAF,[278] endothelin,[279] and calcitonin gene–related peptide.[274,280-283] Elevation of blood flow through vessels with intact endothelium or across cultured endothelial cells results in increased shear stress and increased NO release. Both the pulse frequency and pulse pressure modulate flow-induced NO release.[266,273,275,284-288] Elevated perfusion pressure and shear stress also increase NO release from afferent arterioles.[289]

NO plays a major role in modulation of renal hemodynamics, regulation of medullary perfusion, modulation of the sensitivity of the TGF mechanism, inhibition of tubular sodium reabsorption, modulation of renal sympathetic neural activity, and mediation of pressure natriuresis.[38,290-294] NO dominates integrated renal hyperemic responses to acetylcholine and bradykinin. Renal endothelium–dependent vasodilation is diminished in diabetes due to impaired NO function.[295] Pressure natriuresis in experimental models using stepwise increases of renal perfusion pressure increases NO release, which exerts direct tubular effects to promote sodium and water excretion.[296,297] Tubular epithelial cells are capable of releasing NO, but during increased medullary flow, the vasa recta may be a primary source of the NO, as suggested by the fact that flow-dependent increases of NO also occur, even during microperfusion of isolated outer medullary vasa recta.[298]

There are important interactions among NO, Ang II, and renal nerves in the control of renal function and blood pressure.[299] Nonselective NOS inhibition using competitive inhibitors of NO decreases RPF and results in a smaller reduction in GFR.[172,300,301] These effects are prevented by the simultaneous administration of excess L-arginine, the NOS substrate.[300] Selective inhibition of neuronal NOS (nNOS or NOS1), which is found in the thick ascending limb of the loop of Henle, the macula densa, and efferent arterioles,[263,292] decreases GFR without affecting blood pressure or RBF.[302] Because eNOS is found in the endothelium of renal blood vessels, including the afferent and efferent arterioles and glomerular capillary endothelial cells,[263] differences in the effects of generalized NOS inhibition versus specific inhibition of nNOS on NO formation and RBF appear to be related to the distinct distribution of eNOS versus nNOS in the kidney. Both acute and chronic inhibition of NO production result in systemic and glomerular capillary hypertension, an increase in preglomerular (R_A) and efferent arteriolar (R_E) resistance, a decrease in K_f, and decreases in single-nephron plasma flow and GFR.[303-307]

As shown in Fig. 3.23, acute administration of pressor doses of a blocker of NO production results in a decline in SNGFR, Q_A, and K_f and increases in both preglomerular and efferent arteriolar resistances. Administration of nonpressor

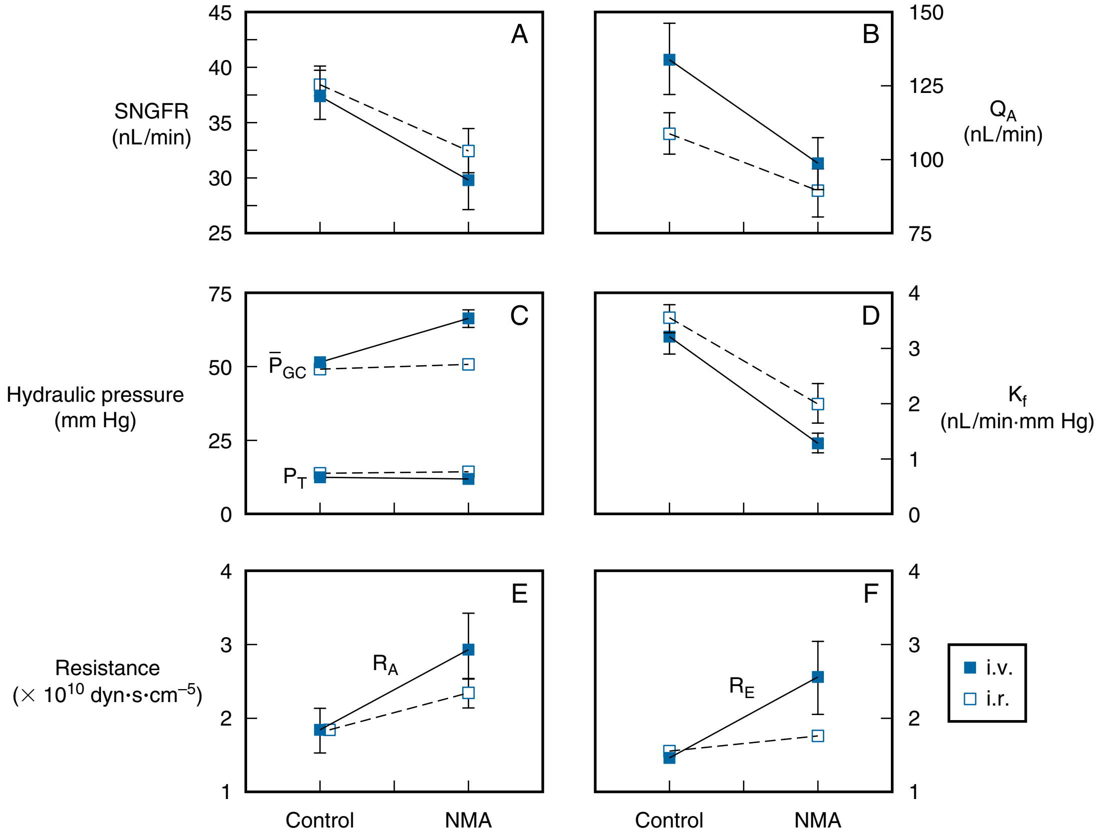

Fig. 3.23 (A–F) Role of nitric oxide in the control of glomerular filtration dynamics. Studies were performed in euvolemic Munich-Wistar rats receiving intravenous pressor doses of the nonselective nitric oxide synthase (NOS) blocker, *N*-monomethyl-L-arginine (*NMA;* i.v., *filled squares*), or nonpressor doses of NMA at the origin of the renal artery (i.r., *open squares*). K_f, Filtration coefficient; P_{GC}, pressure in the glomerular capillaries; P_T, pressure in the tubules; Q_A, glomerular plasma flow rate; R_A, preglomerular arteriolar resistance; R_E, efferent arteriolar resistance; *SNGFR,* single-nephron glomerular filtration rate. (Data [mean ± SE] obtained from Deng A, Baylis C. Locally produced NO controls preglomerular resistance and filtration coefficient. *Am J Physiol*. 1993;264:F212–F215.)

doses of the inhibitor of NO formation through the renal artery yielded an increase in preglomerular resistance and a decrease in SNGFR and K_f but no effect on efferent resistance.[304] These studies suggest that the cortical afferent, but not efferent, arterioles are under tonic control by NO. However, others have found that the renal artery, arcuate and interlobular arteries, and afferent and efferent arterioles all produce NO and constrict in response to inhibition of endogenous NO production.[151,176,240,264,308-311] In agreement with this finding, investigators[34,309] have reported that NO dilates both efferent and afferent arterioles in perfused juxtamedullary nephrons. Interestingly, the modulatory influence of nNOS on afferent arteriolar tone is dependent on the maintenance of distal tubular fluid, indicative of a critical interaction with the TGF mechanism.[311]

Controversy exists regarding the role of the RAS in the genesis of the increase in vascular resistance that follows blockade of NOS. Studies of in vitro perfused nephrons[309] and of anesthetized rats in vivo[312] suggest that the increase in renal vascular resistance that follows NOS blockade is blunted when Ang II formation or receptor binding is blocked. NO inhibits renin release, whereas acute Ang II infusion increases cortical NOS activity and protein expression, and chronic Ang II infusion increases mRNA levels for eNOS and nNOS.[312,313] Ang II increases NO production in isolated perfused afferent arterioles via activation of the AT1 Ang II receptors.[314] In contrast, nonselective NOS inhibition increases renal oxygen consumption, independently of Ang II.[315] Additionally, inhibition of NOS in conscious rats had similar effects on renal hemodynamics in the intact and Ang II–blocked state.[316] This suggests that the vasoconstrictor response to NOS blockade is not mediated by Ang II. Further studies[317] have shown that when the Ang II levels are acutely raised by the infusion of Ang II, acute NO blockade amplifies the renal vasoconstrictor actions of Ang II. An observation in agreement with this finding is that intrarenal inhibition of NO enhances Ang II–induced afferent, but not efferent, arteriolar vasoconstriction.[151,318] In the juxtaglomerular nephron, however, blockade of nNOS enhanced efferent but not afferent arteriolar responsiveness to Ang II.[311] These data indicate that NO modulates the vasoconstrictor effects of Ang II on glomerular arterioles in vivo, perhaps blunting Ang II's vasoconstrictor response in the afferent arteriole, with some results showing similar responses on the efferent arteriole.

Effects of Heme Oxygenase and Carbon Monoxide on Renal Function

Heme is degraded by heme oxygenase (HO) enzymes (HO-1 and HO-2) producing carbon monoxide (CO), biliverdin, and bilirubin and by the release of free iron.[2,38,319-322] Induction of HO-1 with hemin in anesthetized rats resulted in significant increases in RBF and GFR, reduced renal vascular resistance, and an increase in sodium excretion in untreated control rats without affecting blood pressure. Furthermore, autoregulatory responses to acute Ang II infusion were blunted, and these studies indicated a vasodilatory influence of HO-1 induction with increased CO production.[320] When a heme oxygenase inhibitor was administered either alone or to rats receiving an NOS inhibitor—N(ω)-nitro-L-arginine methyl ester (L-NAME)—for 4 days, blockade of HO in control rats decreased endogenous CO, HO-1 levels, urine volume, and sodium excretion but did not affect AP, RBF, or GFR. An increase in plasma renin activity was observed in untreated rats but not in L-NAME–treated rats, indicating that the effects on urine volume and sodium excretion are associated, even when NO was inhibited. This suggests that inhibition of HO promotes water and sodium excretion by a direct tubular action independently of renal hemodynamics or the NO system.[321]

Inhibition of renal medullary HO activity and CO production decreases medullary blood flow and sodium excretion; the abundance of both the HO-1 and HO-2 isoforms of HO are higher in the inner medulla and lower in the cortex.[323] Inhibition of HO significantly reduces renal medullary cGMP concentrations when infused into the renal medullary interstitial space. These results suggest that both HO-1 and HO-2 are highly expressed in the renal medulla and that HO and its products play a key role in maintaining the constancy of blood flow to the renal medulla; cGMP may mediate the vasodilator effect of HO and CO in the renal medullary circulation.[323] In anesthetized rats, increases in renal perfusion pressure increased CO concentrations in the renal medulla. An HO inhibitor reduced HO activity and pressure-dependent increases in CO in the medulla and blunted pressure natriuresis.[324] In conscious rats fed a normal-sodium diet, chronic infusion of an HO inhibitor into the renal medulla increased mean AP. When rats were placed on a high-salt diet, inhibition of HO activity caused a further increase in AP. Thus renal medullary HO activity plays a key role in the control of arterial blood pressure and the control of pressure natriuresis.[324]

A CO-releasing molecule (CORM-A1) increased CO and RBF, with comparable results obtained from infusion of the vasodilator acetylcholine.[325] Pretreatment with an inhibitor of guanylate cyclase to block acetylcholine reduced the increase in RBF by CORM-A1. In isolated vasoconstricted renal interlobular arteries, CORM-A1–induced vasodilation was attenuated with the guanylate cyclase inhibitor. Inhibition of calcium-activated potassium channels (K_{Ca}) with iberiotoxin blocked CORM-A1 vasodilation. Thus CO released from CORM-AI increases RBF and decreases vascular resistance by activating guanylate cyclase and opening K_{Ca} channels.[325]

The vascular effects of HO may be related to CO synthesis and are affected by NO release linked to the HO-CO system. Administration of a CO donor into the renal artery of rats increased RBF, GFR urinary cGMP excretion, and blood carboxyhemoglobin levels.[326] Inhibition of HO-induced acute renal failure, with decreases in RBF, GFR, and cGMP excretion. These effects were nearly eliminated by the addition of a CO donor, which also decreased renal cortical NO concentration, urinary excretion of nitrates and nitrites, and urinary cGMP excretion and increased blood carboxyhemoglobin levels. Inhibition of renal HO resulted in acute renal failure, characterized by large decreases in RBF and GFR. Supplementing HO inhibition with CO donor administration reversed the effects of HO inhibition on RBF and GFR, indicating that the deleterious effects of HO on RBF and GFR were caused by the inhibition of CO. HO inhibition also decreased cortical NO concentration and increased urinary nitrate and/or nitrite excretion of the HO-CO system, whereas a CO donor increased renal NO levels and decreased nitrate and nitrite excretion. These results suggest that changes in NO release contribute to the renal effects of the HO-CO system.[326]

Administration of heme decreases vascular resistance and increases RBF and sodium excretion, excretion of 6-keto-PGF1α, and the concentration of CO in renal cortical microdialysate. Pretreatment with an inhibitor of HO blunted heme-induced renal vasodilation and increased RBF. Pretreatment with sodium meclofenamate blunted the renal vasodilatory effect of heme, suggesting that heme-induced renal vasodilation is cyclooxygenase dependent, yielding increased synthesis of PGI2.

Hydrogen Sulfide

H_2S, an endogenous bioactive gas synthesized in nearly all organs, also contributes to the regulation of kidney function. H_2S generation by kidney cells is reduced in acute and chronic disease states, and H_2S donors ameliorate injury,[327] but under some conditions, H_2S may lead to kidney injury.[328] H_2S is produced by cystathionine β-synthase (CBS) and cystathionine γ-lyase (CGL) by the transsulfuration of homocysteine.[329,330] Incubation of renal tissue homogenates with L-cysteine as a substrate yields H_2S. This response was prevented by inhibitors of both CBS and CGL in combination, whereas either inhibitor alone induced only a small decrease in H_2S.[329]

Intrarenal infusion of a donor of H_2S (NaHS) increased RBF and GFR, as well as urinary sodium and potassium excretion. Infusion of L-cysteine also increases endogenous H_2S production.[329] Simultaneous infusion of both an inhibitor of CBS and CGL to decrease H_2S production decreased GFR and sodium and potassium excretion, but either inhibitor alone did not affect these renal functions.[329] H_2S causes endothelium-dependent/cytochrome P450–dependent vasodilation and vascular smooth muscle hyperpolarization of small arterial vessels, increasing ryanodine-mediated Ca^{2+} release through the activation of large-conductance calcium-activated potassium channels, causing membrane hyperpolarization and vasodilation.[331] NaHS-induced vasorelaxation was reduced by removal of the endothelium and by inhibitors of either NO or cGMP production.[332] H_2S also relaxes smooth muscle by activating ATP-sensitive potassium channels.[333]

Increases in TGF-β1 are associated with the development of tubulointerstitial fibrosis and glomerular sclerosis and are mediated, at least in part, by Ang II. Ang II– and transforming growth factor β1 (TGF-β1)–induced renal tubular epithelial-mesenchymal transition (EMT) plays a pivotal role leading to renal sclerosis. Ang II stimulates EMT in renal tubular epithelial cells by increasing the level of α-smooth actin and decreasing E-cadherin.[334] This effect was blocked by a TGF-β receptor kinase inhibitor.

Ang II stimulated TGF-β activation and exogenous TGF-β1–induced EMT. The H_2S donor NaHS blocked the promotion of EMT by Ang II and TGF-β1 and reduced TGF-β activity. H_2S cleaves the disulfide bond in dimeric active TGF-β1, promoting the formation of inactive TGF-β1 monomer.[334] These results have suggested the potential to treat glomerular sclerosis and tubulointerstitial fibrosis by stimulating H_2S or using another means to form the inactive TGF-β1 monomer.

REACTIVE OXYGEN SPECIES

Reactive oxygen species (ROS) are products of a one-electron reduction of dioxygen (oxygen gas, O_2) to form the anionic form of O_2, superoxide, O_2^-. Superoxide is generated by the catalytic actions of oxidative enzymes, such as nicotinamide adenine dinucleotide (NADH)/reduced NADPH oxidase (NOX) and cytochrome oxidase.[2,38] Superoxide is toxic, and organisms creating O_2^- have developed isoforms of superoxide dismutase (SOD), which catalyzes the conversion of superoxide to hydrogen peroxide (H_2O_2). Other enzymes degrade H_2O_2, protecting against the deleterious actions of this ROS.[2,38] ROS are produced in the kidney by most cell types including endothelial cells, epithelial cells, vascular smooth muscle cells, mesangial cells, and podocytes and have deleterious effects on the kidney vasculature.[335]

Normally, oxidative and antioxidative enzymes in the kidney yield a balanced production of NO and the superoxide anion. ROS are formed in the arteries, arterioles, glomeruli, juxtaglomerular apparatus, and other nephron segments, and the oxidases NOX 1, 2, and 4, NOS, and COX are also formed in the kidney.[2,38] In the renal vasculature, NOX 1 and NOX 2 produce O_2^-, whereas NOX 4 in epithelial cells produces H_2O_2. Superoxide dismutase converts superoxide to H_2O_2; catalase and glutathione peroxidase degrade H_2O_2.

Stimulants of O_2^- production include Ang II,[336-338] endothelin,[339] norepinephrine,[2] TGF-β1,[340] and stretch of vascular walls by increased intravascular pressure.[341,342] Ang II activates NADPH oxidases in afferent arterioles to form O_2^-, leading to calcium release from intracellular stores by activation of the inositol trisphosphate receptor (IP3R).[336] Superoxide also enhances calcium entry pathways into afferent arterioles through L-type channels by membrane depolarization.[343] NOX2 NADPH oxidase is activated by Ang II, promoting the generation of ROS, which scavenges NO and causes subsequent NO deficiency.[337] O_2^- and H_2O_2 activate different signaling pathways in vascular smooth muscle cells linked to discrete membrane channels, with opposite effects on membrane potential and voltage-operated Ca^{2+} channels, and therefore have opposite effects on myogenic contractions.[341]

Increased perfusion pressure and Ang II increase O_2^- production and increase myogenic responses in arterioles of superoxide gene–deleted mice compared with controls.[344] In the macula densa of blood-perfused juxtamedullary nephrons, O_2^- was undetectable in control normotensive mice but markedly elevated in Ang II-induced hypertensive animals. NO was found in the macula densa of control mice but was undetectable in the macula densa of hypertensive animals.[345] These data suggest that under normal conditions, NO generated in the macula densa reduces TGF sensitivity, but in Ang II-induced hypertension, the TGF response is augmented by O_2^- generated by the macula densa.[345] Increased perfusion pressure causes vascular O_2^- production from NADPH oxidase, enhancing myogenic contractions independently of NO, whereas H_2O_2 impairs pressure-induced contractions but is not involved in the normal myogenic response.[346]

In the remnant kidney model, COX-2 is induced, leading to activation of thromboxane receptors (TP-Rs), which enhance ET-1, ROS generation, and contractions.[339] Compared with controls, diabetic mouse afferent arterioles also have increased production of O_2^- and H_2O_2 and enhanced responses to ET-1. These responses are accompanied by reduced protein expression and activities for catalase and superoxide dismutase-2. ET-1 further increases O_2^-, whereas H_2O_2 is unchanged by ET-1. Increased ROS in diabetes (notably H_2O_2) contributes to the enhanced arteriolar responses to ET-1.[347] TGF-β1, a growth factor involved in glomerular and tubular injury in diabetes, blocks autoregulation of afferent arterioles, an effect prevented with a ROS scavenger or an NADPH oxidase inhibitor. In smooth muscle cells, TGF-β1 stimulated ROS formation that was inhibited by NADPH oxidase inhibitors.[340]

In afferent arterioles from rats with spontaneous hypertension, pressure-induced increases in ROS were four times greater in SHR than in WKY rats. Both a scavenger of O_2^- and a NOX2-based (NADPH oxidase) inhibitor attenuated pressure-induced constriction in SHR vessels but not in WKY. Thus NOX2-derived O_2^- may contribute to an enhanced myogenic response in SHR afferent arterioles.[348] Of note, arterioles from rats with ischemia-reperfusion injury had a 38% increase in H_2O_2, which could act to buffer the effect of Ang II and be a protective mechanism.[349]

Endothelin

Endothelin is a potent vasoconstrictor agent derived primarily from vascular endothelial cells.[350] There are three distinct genes for endothelin, each encoding distinct 21–amino acid isopeptides, ET-1, ET-2, and ET-3.[350,352] Proteolytic cleavage of a 212–amino acid preproendothelin by furin yields a 38- to 40-amino acid proendothelin, which in turn is cleaved by endothelin-converting enzyme to yield endothelin peptides.[353,354] ET-1, the primary endothelin produced in the kidney, is formed in arcuate arteries and veins, interlobular arteries, afferent and efferent arterioles, glomerular capillary endothelial cells, glomerular epithelial cells, and glomerular mesangial cells.[355-366] ET-1 acts in an autocrine or paracrine fashion, or both,[367] to alter a variety of biologic processes in these cells. Endothelins are mostly vasoconstrictors, and the renal vasculature is highly sensitive to these agents.[368] Once released from endothelial cells, endothelins bind to specific receptors on vascular smooth muscle. ET_A receptors bind both ET-1 and ET-2.[367,369-372] ET_B receptors are expressed in the glomerulus on mesangial cells and podocytes and have equal affinity for ET-1, ET-2, and ET-3.[371,372] There are two subtypes of ET_B receptors, the ET_{B1} linked to vasodilation and the ET_{B2} linked to vasoconstriction.[373] An endothelin-specific protease modulates endothelin levels in the kidney.[374-376]

Endothelin production is stimulated by physical factors, including shear stress and vascular stretch.[377,378] Various hormones, growth factors, and vasoactive peptides increase endothelin production, including TGF-β, platelet-derived growth factor, tumor necrosis factor-α, Ang II, arginine vasopressin, insulin, bradykinin, thromboxane A2, and thrombin.[355,359,361,365,379-382] Endothelin production is inhibited by atrial and brain natriuretic peptides acting through a cGMP-dependent process[374,383] and by factors that increase intracellular cAMP and protein kinase A activation, such as β-adrenergic agonists.[361] ATP-binding renal purinergic (P2) receptors may also regulate ET-1 production.[384]

Intravenous infusion of ET-1 induces a marked, prolonged pressor response[350,385] accompanied by increases in preglomerular and efferent arteriolar resistances and a decrease in RBF and GFR, without changes in fractional Na excretion.[385] As shown in Fig. 3.24, infusion of subpressor doses of ET-1 decreases SNGFR, Q_A, and whole-kidney RBF and GFR,[386-390] accompanied by increases in preglomerular and postglomerular resistances and filtration fraction.[386,388,391] Vasoconstriction of afferent and efferent arterioles by endothelin has been confirmed in the split hydronephrotic rat kidney preparation[392,393] and in isolated perfused arterioles.[308,394,395] Endothelin also causes mesangial cell contraction and reduces K_f.[352,396] The vasoconstrictor effects of the endothelins can be modulated by several factors,[370,397] including NO,[308,398] bradykinin,[399] prostaglandin E2,[400] and prostacyclin.[400,401] The endothelin pathways demonstrate significant sexual dimorphisms that may affect the progression of renal disease and treatment choices.[402]

The ET_A and ET_B receptors have been cloned and characterized.[372,403,404] ET_A receptors are abundant on vascular smooth muscle, have a high affinity for ET-1 and play a prominent role in the pressor response to endothelin.[405] ET_B receptors are present on endothelial cells, where they may mediate NO release and relaxation.[404] Both ET_A and ET_B receptors are expressed in the media of interlobular arteries and afferent and efferent arterioles. Only ET_A receptors are present on vascular smooth muscle cells of interlobar and arcuate arteries.[406] ET_B receptors are on peritubular and glomerular capillaries, as well as the vasa recta endothelium.[406] ET_A receptors are evident on glomerular mesangial cells and pericytes of descending vasa recta bundles.[406]

ET_A receptor antagonists may be useful for patients with diabetic nephropathy[407] as they reduce albuminuria, although the albuminuria returns on cessation of the drug. It has been used alone and in conjunction with renin-angiotensin blockade.[408,409] However, ET_A receptor antagonists have been associated with edema and heart failure, so patients already manifesting these conditions should be excluded from this type of medication.[410]

Endothelin stimulates the production of vasodilatory prostaglandins,[389,398,401,411,412] yielding a feedback loop to dampen the vasoconstrictor effects of endothelin. ET-1, ET-2, and ET-3 also stimulate NO production in the arterioles and glomerular mesangium via activation of the ET_B receptor.[277,279,308,398,413] There is a dynamic relationship between NO and endothelin effects, so ET_A blockade or inhibition of endothelin-converting enzyme leads to increased renal resistance caused by NO inhibition.[414,415]

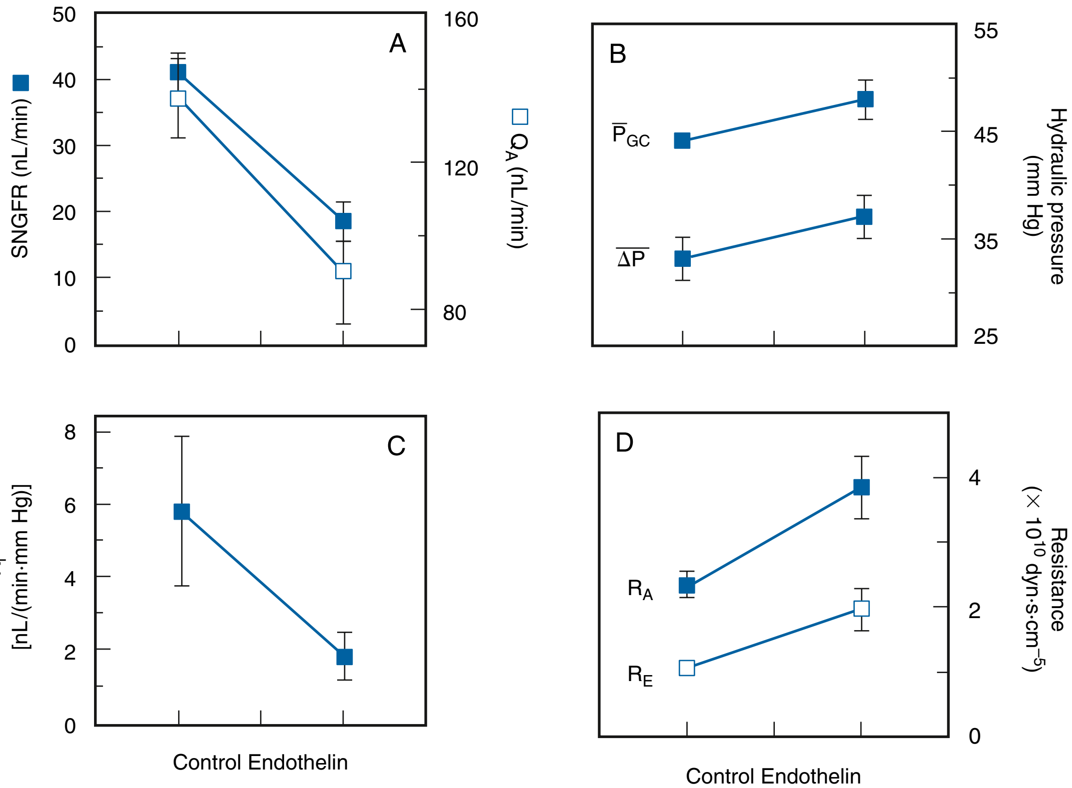

Fig. 3.24 (A–D) Effects of intravenous administration of endothelin (subpressor dose) on glomerular dynamics. *K_f*, Filtration coefficient; *ΔP*, mean transcapillary pressure gradient; *P_{GC}*, mean glomerular capillary pressure; *Q_A*, single-nephron glomerular plasma flow; *R_A*, afferent arteriolar resistance; *R_E*, efferent arteriolar resistance; *SNGFR,* single-nephron glomerular filtration rate. (Data [mean ± SE] obtained in Munich-Wistar rats from Badr KF, et al. Mesangial cell, glomerular and renal vascular responses to endothelin in the rat kidney. Elucidation of signal transduction pathways. *J Clin Invest*. 1989;83(1):336–342.)

The vasoconstrictive effects of Ang II may be mediated, in part, by stimulation of ET-1 production, which acts on ET_A receptors to produce vasoconstriction.[379,382] Chronic administration of Ang II reduces RBF, an effect reduced by a mixed ET_A-ET_B receptor antagonist, suggesting that endothelin contributes to the renal vasoconstrictive effects of Ang II.[379]

Blood flow through the renal medulla is influenced by ET-1, as well as Ang II, norepinephrine, nitric oxide, and vasodilatory prostaglandins. Medullary vasodilation measured by laser Doppler techniques was seen to occur at low doses of endothelin when cortical blood flow was decreased.[416] An ET_A receptor antagonist blocked cortical vasoconstriction by ET-1 but failed to prevent medullary dilation. The endothelin-induced medullary vasodilation was blocked by an $ET_{A/B}$ receptor antagonist and was mimicked by an ET_B receptor agonist. Inhibition of NO completely blocked the endothelin-induced vasodilation of medullary blood flow, and inhibition of prostaglandins attenuated the response. These results indicate that endothelin causes cortical vasoconstriction mediated by ET_A receptors, whereas activation of ET_B receptors causes medullary vasodilation mediated by the release of NO.[416]

Most regions of the vasa recta are covered by pericytes capable of vasoconstriction. The role of these pericytes in controlling medullary blood flow has been examined with confocal microscopy, and pericyte-mediated vasoconstriction and vasodilation were visualized. Ang II, endothelin, and norepinephrine all caused vasoconstriction at pericyte locations.[417] These effects were attenuated by an NO donor and enhanced with inhibitors of NO production or inhibition of prostaglandin release by nonselective cyclooxygenase inhibition with indomethacin. Because of the narrow diameter of the vasa recta (normally, ~10 μm), constriction of pericytes can cause impairment of the movement of red cells and hence blood flow through the medulla. These results suggest an important role for pericytes in the control of the medullary circulation.[417]

ROLE OF THE RENIN-ANGIOTENSIN SYSTEM IN THE CONTROL OF RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

The RAS exerts major autocrine, paracrine, and endocrine functions regulating RBF and GFR. As presented in detail in several recent reviews,[2,38,61,90] renin is a proteolytic enzyme synthesized, stored, and released from the kidney. It is also synthesized in the liver. In the kidney, it is synthesized and secreted primarily by the granulated epithelioid cells of the JGA adjacent to the terminal portion of the afferent arteriole; renin is also formed in the proximal tubules and in the principal cells of the connecting tubule and collecting duct.[2,38,418-420] Renin release from the kidneys is stimulated by a decrease in sodium intake, a reduction in extracellular fluid volume (ECFV) and blood volume, a decrease in arterial blood pressure, and increased sympathetic nerve activity.[421] Renin cleaves a decapeptide, angiotensin I (Ang I) from angiotensinogen, a glycoprotein formed in the proximal tubules of the kidney[420] and the liver. Circulating angiotensinogen is present in the α_2-globulin fraction of plasma. Subsequent conversion of Ang I by ACE, identical to kininase II, yields the octapeptide Ang II. ACE is present in many tissues including lung. In the kidney, it is bound to the luminal sides of endothelial cells of blood vessels and tubular cells, including the brush border of the proximal tubule. All components needed for the production and degradation (the latter by angiotensinase A) of Ang II are present in the immediate region of the juxtaglomerular region of the nephron, allowing direct local regulation of glomerular blood flow and filtration rate.[38,61]

Ang II is a potent vasoconstrictor, and numerous studies have demonstrated that preglomerular vessels, including the arcuate arteries, interlobular arteries, and afferent arterioles, as well as the postglomerular efferent arterioles, constrict in response to exogenous and endogenous Ang II.[151,181,310,395,422,423]

Some studies have indicated that efferent arterioles have a greater sensitivity to Ang II, whereas others have shown similar effects on both afferent and efferent arterioles.[38,151,310,395,423] Fig. 3.25 shows the effects of Ang II on diameters in these vessels; both L-type and T-type Ca^+ channels are involved in the afferent arteriolar responses to Ang II while the T-type Ca^+ channels predominate at the efferent arterioles.[424,425] In addition to constricting vascular smooth muscle cells, Ang II stimulates various transporters along the nephron, increases myocardial contractility, stimulates aldosterone release, increases salt appetite and thirst, and helps regulate sodium transport by the kidney tubules and intestine.[426] The overall effect of Ang II is to minimize renal fluid and sodium losses and maintain ECFV and arterial blood pressure.[421]

There are two major classes of Ang II receptors, AT1 and AT2, but the hypertensinogenic and vasoconstrictive actions of Ang II are primarily due to the AT1 receptor, which is widely distributed throughout all segments of the renal microvasculature in the cortical and medullary circulatory beds and are also present in the glomeruli, including mesangial cells, glomerular capillary endothelial cells, and podocytes.[38,90,427] In rodents, afferent arterioles have both AT1a and AT1b receptors, whereas efferent arterioles have only AT1a receptors.[427,428] AT1 receptors are also present in the proximal and distal tubules, loop of Henle and macula densa cells, cortical and medullary collecting ducts, glomerular podocytes, and mesangial cells.[426]

The vasoconstrictor effects of Ang II are blunted by the endogenous production of vasodilators including NO, cyclooxygenase, and cytochrome P450 epoxygenase metabolites in the afferent but not efferent arterioles.[147,151,310,318,429-432] Ang II-simulated release of NO in the afferent arterioles occurs through activation of the AT1 receptors.[314,433] Ang II increases the production of prostaglandins (both PGE2 and PGI2) in afferent arteriolar smooth muscle cells, and PGE2, PGI2, and cAMP all blunt Ang II–induced calcium entry into these cells,[430] potentially explaining, at least in part, the different effects of Ang II on vasoconstriction of the afferent and efferent arterioles.[429,430] In contrast to effects on the afferent arteriole, PGE2 has no effect on Ang II-induced vasoconstriction of the efferent arteriole.[147] The effects of PGE2 on Ang II–induced vasoconstriction of the afferent arteriole are concentration dependent, with low concentrations acting as a vasodilator via interaction with prostaglandin EP4 receptors and high concentrations of

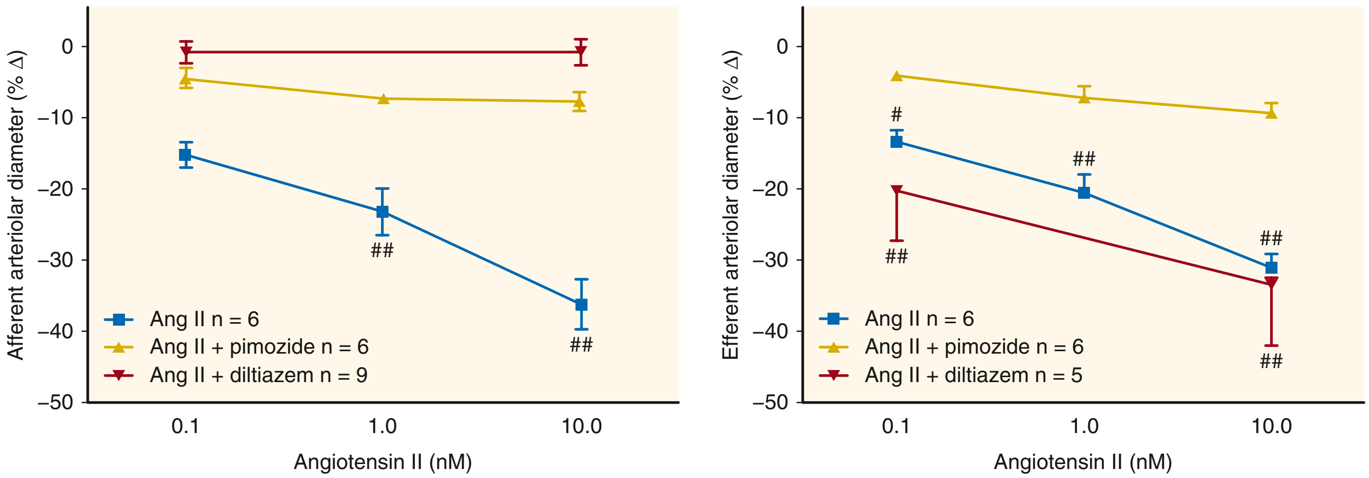

Fig. 3.25 Role of L- and T-type Ca^{2+} channels in angiotensin II (Ang II)–mediated afferent and efferent arteriolar vasoconstriction. Ang II constricts both afferent and efferent arterioles. The vasoconstrictor effects of Ang II on afferent arterioles are blocked by both L-type and T-type Ca^{2+} channel blockers. In contrast, the efferent vasoconstrictor effects of Ang II are not blocked by L-type Ca^{2+} channel blockers but are blocked by T-type Ca^{2+} channel blockers.[35,526,527] (From Navar LG, Arendshorst WJ, Pallone TL, et al. The renal microcirculation. In: Tuma RF, Duran WN, Ley K, eds. *Handbook of Physiology: Microcirculation.* Vol 2. Cambridge, MA: Academic Press; 2008:550–683. Carmines PK, Navar LG. Disparate effects of Ca channel blockade on afferent and efferent arteriolar responses to Ang II. *Am J Physiol Renal Physiol*. 1989;256(6 Pt 2):F1015–F1020; Feng MG, Navar LG. Angiotensin II-mediated constriction of afferent and efferent arterioles involves T-type Ca^{2+} channel activation. *Am J Nephrol.* 2004;24(6):641–648.)

PGE2 acting on prostaglandin EP3 receptors to restore the Ang II effects in that segment.[147] Ang II infusion alone decreases RBF, with lesser effects on GFR resulting in an increase in filtration fraction.[38] However, when combined with cyclooxygenase inhibition, Ang II causes marked reductions in SNGFR and Q_A, suggesting an important role for endogenous vasodilatory prostaglandins in ameliorating the vasoconstrictor effects of Ang II.[434] Because Ang II increases renal production of vasodilatory prostaglandins, this serves as a feedback loop to modulate the vasoconstrictor effects on Ang II under chronic conditions when the RAS is stimulated.[90]

Ang II decreases K_f[434] and contracts mesangial cells.[435] One possible cause for the changes in K_f, is that contraction of the mesangial cells reduces effective filtration area by blocking flow through some glomerular capillaries. Alternatively, Ang II might decrease hydraulic conductivity, rather than or as well as reducing the surface area available for filtration, thereby reducing K_f.[436] Glomerular epithelial cells possess both AT1 and AT2 receptors and respond to Ang II by increasing cAMP production, suggesting a possible role for these cells in reducing K_f.[437] Alterations in epithelial structure or the size of the filtration slits, however, have not been detected following infusion of Ang II at a dose sufficient to decrease GFR and K_f.[438]

The vasoconstrictive effect of Ang II on glomerular mesangial cells is markedly reduced by NO. Release of NO from endothelial cells stimulated by bradykinin increases cGMP production in co-incubated mesangial cells. Comparable results were obtained when mesangial cells were incubated with NO alone.[269] Ang II alone caused constriction of the mesangial cells, but this effect was largely eliminated when the cells were co-incubated with both Ang II and NO. These studies suggest that local NO production from endothelial cells can modify the effects of Ang II on glomerular mesangial cells.[269] Glomerular epithelial cells contain Ang II receptors and, together with mesangial cells, may play an important role in Ang II-mediated control of the glomerular filtration barrier.[437]

Blockade of AT1 receptors caused a dose-dependent dilation when Ang II was added to the lumen or bath of isolated afferent arterioles preconstricted by norepinephrine. This effect was blocked by pretreatment with an AT2 receptor antagonist, indicating that activation of the AT2 receptors vasodilates afferent arterioles. Disruption of the endothelium or simultaneous inhibition of the cytochrome P450 pathway also abolished the vasodilation by Ang II. Thus activation of AT2 receptors may cause endothelium-dependent vasodilation via the cytochrome P450 pathway, counteracting the vasoconstrictor effects of Ang II at AT1 receptors.[439,440]

As indicated, Ang II produces prohypertensive and renal vasoconstrictor effects via AT1 receptors, whereas activation of AT2 receptors results in modest vasodilation.[38,440] Several metabolic Ang II fragments, once believed to be inactive, have now been shown to also have effects on the kidney, often opposing the actions of Ang II. The related peptide, Ang 1-7, induces vasodilation of preconstricted renal arterioles.[441] These effects occur independently of binding to AT1 or AT2 receptors and involve activation of the G protein–coupled Mas receptor,[442] which is expressed on the afferent arteriole.[443] In addition, an isoform of ACE, known as “ACE2,”[444,445] is involved in the formation of Ang 1-7.[446] The balance between opposing actions of the vasoconstrictor peptide, Ang II, and the vasodilator peptide, Ang 1-7, may be influenced by the ratio of ACE to ACE2 and AT1 to Mas receptor content in specific vascular regions (and tubular segments) of the kidney. Cardiovascular and renal diseases may involve an imbalance of these peptides, enzymes, or receptors.[443]

Additional mechanisms involved in Ang II-induced hypertension have been proposed. The principal cells of the connecting tubule and collecting duct express prorenin receptors and produce renin, which, coupled with delivery of angiotensinogen from the proximal tubule,[419] allows for the local formation of Ang I.[2,418,440] Luminal ACE in the collecting duct is then available to produce Ang II, which in turn can increase sodium reabsorption by enhancing activity of $E_{Na}C$ and other transporters in the collecting duct, an effect that may be important in the progression of diabetes and Ang II-induced hypertension.[418,419,440,447]

ROLE OF ANGIOTENSIN II IN GLOMERULAR AND TISSUE INJURY

Chronic renal failure is characterized by a progressive decline in renal function, with glomerular and tubular injury and systemic hypertension. An important demonstration of an elevation of glomerular hydraulic pressure in kidney disease was in a model of nephrotoxic serum nephritis (NSN),[448] where Ang II is thought to have a key role in the renal complications that develop. Intravenous infusion of Ang II increases P_{GC} and ΔP in rats with reduced renal mass.[449] In the remnant kidney model of renal failure, inhibition of Ang II production by an ACE inhibitor lowered P_{GC} and ΔP toward normal and prevented proteinuria and glomerular injury.[450] Over the past 30 years, the efficacy of blockade of the RAS with ACE inhibitors and/or Ang II type 1 receptor antagonists (angiotensin receptor blockers [ARBs]) in limiting or slowing the progression of some types of kidney disease has been established (for further discussion on progression and management of chronic kidney disease, see Chapters 49 and 53).[451] ACE inhibitors and ARBs are also effective in the treatment of spontaneous glomerulosclerosis associated with aging.[452,453] Both ACE inhibitors and ARBs prevented hypertension, limited proteinuria, and ameliorated sclerosis in several experimental models of glomerulonephritis.[454] In one study, ACE inhibition promoted regeneration of new renal tissue.[455] Similarly, in the remnant kidney model of renal disease, chronic treatment with an ACE inhibitor causes reversal of renal injury.[456]

Accumulation of extracellular matrix proteins, including collagen and fibronectin, leads to glomerular and tubular injury.[457] Enhanced production of TGF-β1 in the remnant kidney increases matrix formation by stimulating the production of TGF-β1. Administration of a neutralizing antibody or an Ang II receptor antagonist reduces plasma TGF-β1 levels and blood pressure toward normal.[458] ACE inhibition in the obese Zucker rat reduced proteinuria compared with untreated obese animals, downregulated expression of nephrin, and decreased expression of TGB-β1, collagen IV, and fibronectin.[459] Similarly, in an animal model of myocardial infarction (MI), rats with MI had significant renal injury, and treatment by ACE inhibition exerted beneficial effects.[460]

Ang II induces upregulation and activation of a transcription factor called “sterol-responsive element-binding protein” (SREBP-1), leading to upregulation of TGF-β1 and increased production of matrix collagen and fibronectin.[457] ARB treatment prevented the increase in TGF-β1 formation and the induction of SREBP-1. Inhibition of SREBP-1 prevented Ang II–induced glomerular upregulation of TGF-β1 and matrix synthesis, demonstrating a significant role of the transcription factor in glomerular injury, and indicating that TGF-β1 plays a role in the stimulation of matrix production.[457] SREBP-1 upregulation is observed in diabetic kidneys; SREBP-1 is activated by high glucose levels and mediates profibrogenic responses in primary rat mesangial cells.[461] In these cells, high glucose levels activated SREBP-1, which binds to the TGF-β1 promoter, resulting in TGF-β1 upregulation.[461]

Treatment with an ARB in rats with NSN protected against interstitial inflammation, tubular degeneration, and segmental glomerulosclerosis.[462] NSN rats were treated with pirfenidone (an antifibrotic, antiinflammatory compound first used to treat pulmonary fibrosis) or with pirfenidone plus an ARB. Pirfenidone alone yielded a decrease in proteinuria and in the ARB a slightly greater decrease occurred; the two together had additive effects on proteinuria, interstitial inflammation, tubular degeneration, and segmental glomerulosclerosis.

ARACHIDONIC ACID METABOLITES REGULATING RENAL BLOOD FLOW AND GLOMERULAR FILTRATION RATE

The processing of the essential polyunsaturated fatty acid, linoleic acid, in the liver yields the polyunsaturated fatty acid arachidonic acid, which is stored in membrane phospholipids. Biologically active eicosanoids, C20 metabolites of arachidonic acid, are produced by three primary enzymatic pathways, including the following: cyclooxygenase-generating prostaglandins (PGs) and thromboxanes (TBXs); lipoxygenase (OX) yielding leukotrienes (LTs) and HETEs; and cytochrome P450 pathways synthesizing HETEs and EETs. Arachidonic acid (AA) is released from cell membranes, predominantly by phospholipase A2 (PLA2).[463-467] These eicosanoids regulate the renal microcirculation, in part by activating G protein–coupled receptors in the endothelium and vascular smooth muscles.[464,468]

Prostaglandins

Following the interaction of various hormones and vasoactive substances with their membrane receptors, PLA2 is activated, resulting in the release of AA from the cell membranes. This allows the enzymatic action of cyclooxygenase to process AA into prostaglandins—PGG2 and subsequently PGH2. PGH2 is then converted into a number of biologically active prostaglandins, including PGE2, prostacyclin (PGI2), PGF2α, PGE1, PGD2, and thromboxane (TxA2). Prostaglandins play important roles in healthy and diseased kidneys.[100,469-472] PGE2 receptors are abundant in the kidney, and their effects depend on the location of the receptor subtype and interaction with other active compounds. PGE2 activates at least four receptors, EP1 to EP4. Prostaglandin EP1 receptor activation results in contractile effects in smooth muscle through the Gq alpha protein linkage, whereas EP2 and EP4 are relaxants via the Gs alpha subunit. EP3 is an inhibitory receptor (Gi alpha), decreasing cAMP, which causes vasoconstriction.[38,469,470]

In the kidneys, PGE1, PGI2, and PGE2 are vasodilator prostaglandins that generally increase RPF yet produce little or no increase in GFR and SNGFR, in part due to a decline in K_f.[100,470-472] TxA2 generally results in contraction of the glomerular mesangial cells via the Gq α subunit. During blockade of endogenous prostaglandin production, infusion of PGE2 or PGI2 decreases SNGFR and Q_A, accompanied by an increase in renal vascular resistance (particularly R_E), increases in P_{GC} and ΔP, and a decline in K_f.[473] Additional blockade of Ang II receptors during cyclooxygenase inhibition results in vasodilation in response to PGE2 or PGI2, resulting in a return of SNGFR and Q_A equal to or more than control values, a fall in P_{GC} below control values, and a return of K_f to normal.[473] Thus the renal vasoconstriction induced by exogenous PGE2 or PGI2 appears to be mediated by induction of renin and Ang II production.

Vasodilation at the whole-kidney level resulting from PGI2 infusion during cyclooxygenase and Ang II inhibition has not always been observed.[474] Topical application (but not luminal) of PGE2 to the afferent arteriole increased the vasoconstrictive effect of Ang II and norepinephrine, whereas PGI2 only attenuated norepinephrine-induced vasoconstriction.[475] PGE2 also constricted interlobular arteries, but neither prostaglandin produced vasodilation of vessels preconstricted by Ang II.[475] Indomethacin alone induced vasoconstriction of preglomerular and postglomerular resistance vessels of superficial and juxtamedullary nephrons, indicating that vasodilatory prostaglandins normally modulate endogenous vasoconstrictors.[476] However, there appear to be gender-dependent differences in mechanisms because indomethacin treatment results in net renal vasodilation in female rats.[477] The combination of cyclooxygenase inhibition with an ACE inhibitor caused vasodilation of preglomerular but not postglomerular vessels of the cortical nephrons due to the effects of continued NO production in the preglomerular vessels.[476] Additionally, the PGE2 EP3 receptor regulates COX-2 through a negative feedback loop in the thick ascending limb of the loop of Henle.[478] These data, taken together, indicate that differences in the response to vasoactive prostaglandins between superficial and deep nephrons may be due to prostaglandin receptor subtypes and cyclooxygenase activity.

Leukotrienes and Lipoxins

Leukotrienes are a class of lipid products formed from AA via the lipoxygenase pathway. Leukotrienes that affect glomerular filtration and RBF are leukotriene C4 (LTC4), leukotriene D4 (LTD4), and leukotriene B4 (LTB4). LTC4 and LTD4 are potent vasoconstrictors,[479] whereas LTB4 produces moderate renal vasodilation and an increase in RBF, with no change in GFR in the normal rat.[480] Intravenous infusion of LTC4 increases renal vascular resistance, leading to a decrease in RBF and GFR, as well as a decrease in plasma volume and cardiac output.[481,482] The decline in RBF is mitigated by saralasin, an Ang II receptor antagonist, and indomethacin, an inhibitor of cyclooxygenase, indicating the involvement of Ang II and cyclooxygenase products, as well as a direct effect of LTC4 on the renal resistance vessels.[481] Similarly, LTD4 decreased K_f but increased renal vascular resistance, particularly R_E, a fall in Q_A and SNGFR, and a rise in P_{GC} and ΔP during blockade of Ang II and control of renal perfusion pressure, demonstrating a direct effect of this leukotriene on renal hemodynamics.[483] Leukotrienes often mediate drug-associated nephrotoxicity in some models of kidney disease.[484]

Inflammatory injury also activates the 5-, 12-, and 15-lipoxygenase pathways in neutrophils and platelets to form acyclic eicosanoids called lipoxins (LXs), of which there are two main types, LXA4 and LXB4.[485] LXB4 and 7-*cis*-11-*trans*-LXA4 produce renal vasoconstriction.[484,486] By contrast, intrarenal infusion of LXA4 reduces preglomerular resistance (R_A) without affecting R_E, thereby resulting in an increase in P_{GC} and ΔP.[483] The specific vasodilation of the preglomerular vessels by LXA4 was blocked by cyclooxygenase inhibition, indicating that vasodilatory prostaglandins are responsible for this effect.[483,486] Unique to this compound, LXA4 produced vasodilation while simultaneously causing a reduction in K_f.[486] Because P_{GC}, ΔP, and Q_A were increased, SNGFR also increased.[483] Furthermore, transfection of rat kidney with the 15-lipoxygenase gene suppressed inflammation and preserved function in experimental glomerulonephritis.[487] Lipoxins have also been used to treat acute renal failure in mice.[488,489]

Cytochrome P450

A rapidly growing field of study is that of eicosanoids formed through the cytochrome P450 (CYP450) monooxygenase pathway. AA is metabolized via the CYP450 enzymes to other active metabolites, including EETs, dihydrooxyeicosatetraenoic acids (DiHETEs), and HETEs, which act locally in a paracrine manner.[38] Early researchers demonstrated the role of 20-HETE in regulating renal hemodynamics through direct actions, as well as exerting modulatory actions on the TGF mechanism.[466] In addition to its vascular effects, 20-HETE inhibits sodium transport in the proximal tubule and the thick ascending loop of Henle. 20-HETE enhances myogenic tone and vasoconstricts afferent arterioles, which may be partially mediated through the augmentation of the TGF mechanism.[490] Blockade of 20-HETE formation attenuated vascular responses to Ang II, endothelin, norepinephrine, NO, and CO.[227,490] NO inhibits the production of 20-HETE, which contributes to the vasodilatory effect of NO.[227,466,469,491-493]

In contrast to the contractile responses of 20-HETE, some of the EETs mentioned also formed via the CYP450 pathway are potent vasodilators.[38,494] EETs are formed locally and serve as both autocrine and paracrine factors. EETs are endothelium-derived hyperpolarizing factors that elicit potent vasodilation of the afferent arteriole.[494] These actions are mediated in part by activating large-conductance potassium channels.

NEURAL REGULATION OF RENAL CIRCULATION

The renal vasculature, including the afferent and efferent arterioles, macula densa cells of the distal tubule, and glomerular mesangium are richly innervated.[90,495] Innervation includes renal efferent sympathetic adrenergic nerves[495,496] and renal afferent sensory fibers containing peptides, including calcitonin gene-related peptide (CGRP) and substance P.[90,495] Sympathetic efferent nerves innervate all arterial segments from the main renal artery to the afferent arteriole (including the renin-containing juxtaglomerular cells) and efferent arteriole.[495,496] They play an important role in the regulation of renal hemodynamics, sodium transport, and renin secretion.[497] Afferent nerves containing CGRP and substance P are localized primarily in the main renal artery and interlobar arteries, with some innervation also observed in the arcuate artery, interlobular artery, afferent arteriole, and juxtaglomerular apparatus.[495,496] Peptidergic nerve fibers immunoreactive for neuropeptide Y (NPY), neurotensin, vasoactive intestinal polypeptide, and somatostatin are also present in the kidney.[61] Neuronal NOS neurons have been identified in the kidney.[495] The NOS-containing neuronal stomata are localized in the wall of the renal pelvis, at the renal hilus close to the renal artery, along the interlobar arteries, the arcuate arteries, and extending to the afferent arteriole, supporting their role in the control of RBF.[495] They are also present in nerve bundles that have vasomotor and sensory fibers, suggesting that they modulate renal neural function.[495]

Renal nerve stimulation (RNS) alone increased R_A and R_E, resulting in a decrease in plasma flow and SNGFR, without any effect on K_f.[498] When prostaglandin production was inhibited by indomethacin, however, the same level of RNS produced even greater increases in R_A and R_E, accompanied by large declines in plasma flow and SNGFR and decreases in K_f, P_{GC}, and ΔP.[498] When saralasin was administered as a competitive inhibitor of endogenous Ang II in conjunction with indomethacin, RNS had no effect on K_f, but both R_A and R_E still increased and ΔP was slightly reduced.[498] The release of norepinephrine by RNS enhances Ang II production and results in arteriolar vasoconstriction and reductions in K_f. The increase in Ang II production may then enhance vasodilator prostaglandin production,[498,499] which partially ameliorates the constriction.

Continued vasoconstriction by RNS during the blockade of endogenous prostaglandins and Ang II demonstrates the direct vasoconstrictive effects of norepinephrine and other factors like ATP that are coreleased at the nerve terminals on renal microvessels.[429] Inhibition of NOS results in a decline in SNGFR in normal rats but not in rats with surgical renal denervation, suggesting that NO normally modulates the effects of renal adrenergic activity.[500] However, this modulation does not appear to be related to the sympathetic modulation of renin secretion.[501] Under conditions of renal injury and inflammation, the effects of increased renal nerve activity are exacerbated.[502] The actions of increased sympathetic activity to decrease renal cortical perfusion are attenuated by reductions in ROS.[503]

Renal denervation in animals undergoing acute water deprivation (48-hour duration) or with congestive heart failure increases SNGFR, plasma flow, and K_f.[504] This indicates that the natural activity of the renal nerves in these settings plays an important role in the constriction of the arterioles and reductions in K_f that were observed with water deprivation and in congestive heart failure.[504] The vasoconstrictive effects of the renal nerves in both settings were mediated in part by a stimulatory effect on Ang II release, together with direct vasoconstrictive effects on the preglomerular and postglomerular blood vessels.[504] These studies have demonstrated the important role of the renal nerves in pathophysiologic settings.

Clinical trials have focused on whether catheter-based renal artery denervation reduces blood pressure in patients with resistant hypertension. Some studies have demonstrated sustained reduction in blood pressure in patients who underwent renal denervation; however, the blood pressure response did not differ from the sham-treated patients at 6 months after treatment.[505] More recent clinical trials using improved catheters have provided promising long-term effects.[506]

RENAL HEMODYNAMICS IN THE AGING KIDNEY

Aging is commonly associated with the progressive development of chronic renal disease characterized by global glomerulosclerosis, tubular atrophy, interstitial fibrosis, and arteriosclerosis. For further discussion on aging and the kidney, see Chapter 21. Starting at about 40 to 50 years of age, inulin clearance in humans declines approximately 8 mL/min/1.73 m^2 in each subsequent decade of age.[507] In healthy adult living kidney donors, the prevalence of glomerulosclerosis, followed by local tubular atrophy and interstitial fibrosis, was found to increase progressively from 2.7% for patients aged 18 to 29 years to 73% for patients aged 70 to 77 years.[508] Despite these changes, the age-related decline in GFR was not fully explained by these age-related histologic changes.[508,509] Fig. 3.26 shows the declines in GFR beyond the ages of 40 to 50 years.[90,510-512] Kidney disease associated with aging is generally accompanied by declines in GFR, RPF, and RBF along with an increase in filtration fraction. In one study, the calculated filtration coefficient, K_f, was lower in older than younger subjects (4.9 ± 1.7 nL/[min·mm Hg] in older subjects vs. 7.0 ± 2.9 nL/[min·mm Hg] in younger subjects).[513] However, age-related declines in GFR and RBF do not necessarily occur in everyone.[508,512,514] The reality is that there is a wide range of RBF and GFR in older populations of men and women with many maintaining renal function levels at or close to normal values. In a study of >2069 men and women with ages ranging from 70 to >90 years, average eGFR for men was 65 + 20 (SD) and for women was 65 + 19 mL/min per 1.73 m^2. Thus in disease-free older individuals, GFR is generally adequate to sustain quality of life and may be maintained close to normal (see Fig. 3.26). Another important finding is that the rate of

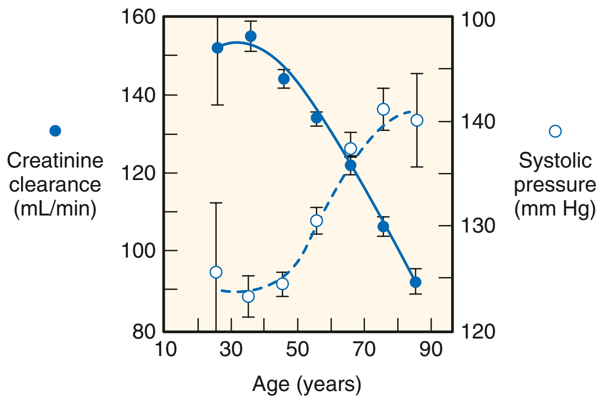

Fig. 3.26 Correlation of aging with systolic blood pressure and creatinine clearance (C_{Cr}) in normal adults. (From Lindeman R, Tobin J, Shock N. Association between blood pressure and the rate of decline in renal function with age. *Kidney Int.* 1984;26: 861–868.)

GFR decline was nonlinear with a decline in the 1-year slope in men and women older than 75, suggesting that the rates of GFR decline between 60 and 75 do not predict the rate of GFR decreases between 75 and 90.[515] As shown in Fig. 3.26, associated with the decline in GFR and RBF, systolic blood pressure increases with age,[513,516] which may contribute to the decline in renal function in individuals with uncontrolled hypertension.

In a cross-sectional and longitudinal study, eGFR data based on using creatinine and cystatin C were determined in 12,000 individuals with ages from 25 to 95 years. Healthy subjects were compared with subjects with diabetes with varying complications. Estimates for aging-dependent declines in eGFR were –0.80 in the general population, –0.79 in healthy subjects, and –1.2 mL/min per 1.73 m^2 in subjects with diabetes. These data confirm that, on average, elderly normal subjects still have lower eGFR values than younger individuals. Greater rates of decline were observed in subjects with diabetes and obesity (–1.32). Within age groups, there was little difference by sex status. More refined age-specific cutoff or lowest acceptable values for healthy individuals were determined for 6 age groups as follows: (<40, 75; 40–50, 70; 50–60, 60; 60–70, 50; 70–80, 40; and <80, 35 mL/min per 1.73 m^2). The important point here is that progressively lower values should be expected throughout the age spectrum while taking into consideration that the variance is large and many older healthy individuals will have much higher levels of GFR.

In this study, there was little indication that the subjects in the normal healthy group had significantly higher blood pressures, thus questioning the dictum that hypertension is a natural consequence of aging. Aging is, however, associated with a higher prevalence of systemic comorbidities. Factors that may contribute to the decline in renal function associated with aging, in addition to hypertension, including the development of diabetes, obesity, oxidative stress due to chronic exposure to oxygen-free radicals, dyslipidemia, excessive exposure to some drugs that reduce renal function, infections, atherosclerotic disease, male gender, due to contribution of androgens to increase fibrosis and mesangial expansion, genetic and racial differences, and environmental factors.[512,517-519] Sclerotic glomeruli seen with aging are smaller than functional glomeruli, leading to a decline in average glomerular size. With the progressive loss of filtering glomeruli, unaffected glomeruli undergo compensatory hypertrophy, which helps preserve GFR.[508] As has been shown in rat studies, following removal of 50% to 80% of total renal mass, the remaining nephrons undergo compensatory hypertrophy, accompanied by increases in SNGFR and glomerular capillary hydraulic pressures.[14,520,521] In the aging human, progressive loss of functioning nephrons and hypertrophy of remaining filtering units also results in glomerular capillary hyperfiltration and hypertension, which increases the vulnerability of the remaining nephrons. The loss of functional glomeruli is reflected by the progressive decrease in cortical volume during aging. Because these alterations use renal reserve capacity, there is a reduced capability of the renal vasculature in older individuals to respond to amino acid infusion, so there is only an increase in GFR and filtration fraction and RPF remains unchanged.[514,522] In contrast, the vasoconstrictive response to Ang II is identical in younger and older people.[523] These data suggest that there is a blunted renal vasodilatory response, but the vasoconstrictive response is maintained in older subjects, indicating that the aged kidney uses its reserve capacity at baseline and is in a state of compensatory renal vasodilation to compensate for the chronic glomerular injury and loss of functional glomeruli.[522] Interestingly, the klotho gene is robustly expressed in the kidneys and is inversely associated with RAS activity and aging. Elevated Ang II levels decrease klotho, and klotho gene transfer reduced the Ang II–induced renal injury and fibrosis. Klotho expression decreases progressively during the aging process, allowing increased Wnt activation associated with podocyte damage and kidney fibrosis. Thus the klotho gene exerts antiinflammatory antifibrotic actions and its decreased expression during the aging process may render the kidneys more susceptible to various aging-associated factors that cause tissue injury.[524]

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QUESTIONS

1. A 21-year-old woman runs a marathon, and her extracellular fluid volume becomes severely decreased. Immediately following the race, she takes double the prescribed dose of ibuprofen (a nonsteroidal antiinflammatory drug) to alleviate the pain of her aching ankle and knee joints. Which one of the following adverse effects is likely to occur in renal circulation?
   a. Increased prostaglandin levels in the kidney
   b. Decreased efferent arteriolar resistance
   c. Increased afferent arteriolar resistance
   d. Increased urinary excretion of salt and water
   e. Increased renal blood flow and GFR

   Answer: c

   Rationale: The decreased extracellular fluid volume stimulates increased renal sympathetic activity and activation of the renin-angiotensin system. This causes direct renal vasoconstriction of both the afferent and efferent arterioles. This is normally counteracted by vasodilator prostanoids such as prostacyclin and PGE_2 that cause afferent vasodilation. However, NSAID drugs block cyclooxygenase and therefore prevent afferent vasodilation, leading to impaired renal blood flow and GFR.

2. A 35-year-old man is diagnosed with severe hypertension due to bilateral renal artery stenosis. An angiotensin receptor antagonist is administered to block the actions of angiotensin II (Ang II) on AT1 receptors. Although blood pressure returns to normal, GFR decreases markedly (by about 50%). In addition to the reduction in blood pressure, which one of the following mechanisms is most responsible for the decrease in GFR?
   a. Increased bradykinin levels in the kidney
   b. Decreased glomerular ultrafiltration coefficient (K_f)
   c. Increased erythropoietin levels in the kidney
   d. Decreased efferent arteriolar resistance
   e. Decreased renin secretion by the juxtaglomerular cells

   Answer: d

   Rationale: Bilateral renal artery stenosis reduces renal perfusion pressure beyond the stenosis, leading to activation of the renin-angiotensin system and increased systemic and intrarenal Ang II levels. The increased Ang II levels increase afferent and efferent arteriolar resistance, but the vasodilator influence of the tubuloglomerular feedback mechanism exerts vasodilator influence on the afferent arterioles, leaving the Ang II influence on the efferent arterioles unopposed. The reduction in blood pressure plus the efferent arteriolar vasodilation due to blockade of the AT receptors cause profound decreases in glomerular pressure and the GFR.

3. Mr. H.I. Stress has not seen a doctor in several years but comes to see you because a screening at a mall showed a blood pressure of 190/100 mm Hg. Records from 10 years previously show that he had a plasma creatinine of 1.0 mg/dL and normal blood pressure. The plasma creatinine concentration is now 4 mg/dL. Which one of the following changes in renal function is illustrated in this patient?
   a. His renal function has been stable.
   b. We cannot make conclusions based on data given.
   c. He has had a 75% reduction in glomerular function.
   d. He has had a 25% reduction in glomerular function.
   e. He has had a 50% reduction in GFR.

   Answer: c

   Rationale: Although not exact, there is a reciprocal relationship between plasma creatinine and GFR such that a fourfold increase in plasma creatinine concentration reflects reduced glomerular functions to of his original value, assuming that the daily production of creatinine has remained unchanged.

4. A 47-year-old man is admitted to the intensive care unit because of severe alcohol withdrawal with symptoms and signs of excessive sympathetic stimulation. The effect of this on renal nerve activity would result in which one of the following?
   a. Decrease in renal blood flow but not GFR
   b. Decrease in both renal blood flow and GFR
   c. Decrease in renal blood flow but increase GFR
   d. Increase in both renal blood flow and GFR due to the increase in arterial pressure
   e. Cause no changes in renal blood flow and GFR because of renal autoregulation

   Answer: b

   Rationale: The renal vasculature has abundant sympathetic innervation to all vascular elements. Thus marked renal sympathetic activity will vasoconstrict both afferent and efferent arterioles causing marked decreases in renal blood flow. Because of the greater amount of preglomerular vascular smooth muscle cells, the preglomerular vasoconstriction will predominate with increased renal sympathetic activity leading to profound decreases in glomerular pressure and thus GFR.

5. A 70-year-old man is seen by his physician for a physical, his first in almost 10 years. He has gained 40 pounds, and his BMI has increased from 29 to 34. His HbA1c is now 9.0, up from 6.0 previously. His serum creatinine has gone from 0.8 to 1.8, suggesting some loss of GFR due to type II diabetes and perhaps an age-related decline in renal function. His blood pressure has also increased from 120/80 to 150/90. The physician prescribes an angiotensin-converting enzyme (ACE) inhibitor to slow the production of angiotensin II (Ang II). Which one of the following responses regarding the effects of Ang II on the kidney is most likely?
   a. The ACE inhibitor will result in a decrease in renal blood flow and GFR.
   b. The reduced intrarenal Ang II will increase sodium reabsorption in the cortical collecting duct.
   c. The ACE inhibitor will lead to reductions in intrarenal Ang II, allowing maintenance of glomerular capillary hydraulic pressure and GFR.
   d. The reduced intrarenal Ang II will decrease production of proliferative factors.
   e. The ACE inhibition will increase intrarenal Ang II.

   Answer: c

   Rationale: Following the conversion of circulating and intrarenal angiotensinogen by renal renin, all components needed to synthesize and degrade Ang II are present in the immediate region of the juxtaglomerular region of the nephron, thereby allowing direct local regulation of glomerular blood flow and filtration rate. Both ACE inhibitors and ARBs prevent hypertension, limit proteinuria, and ameliorate sclerosis in several experimental models of glomerulonephritis.